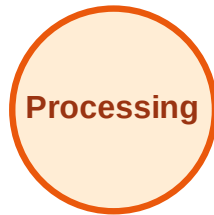


BFS Traversal

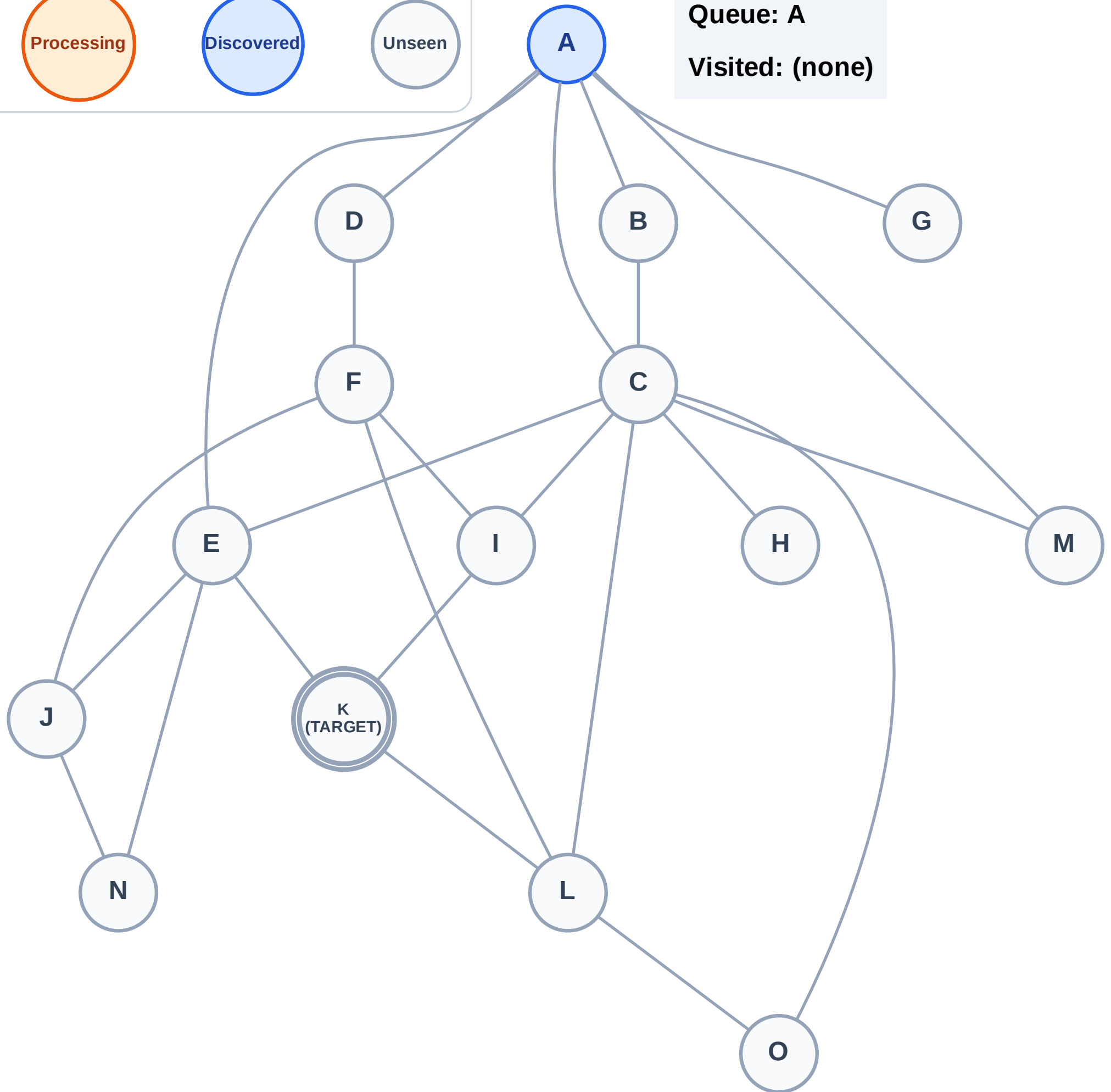
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

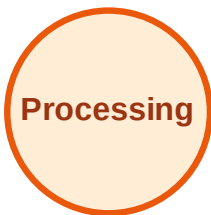
Visited: (none)



BFS Traversal

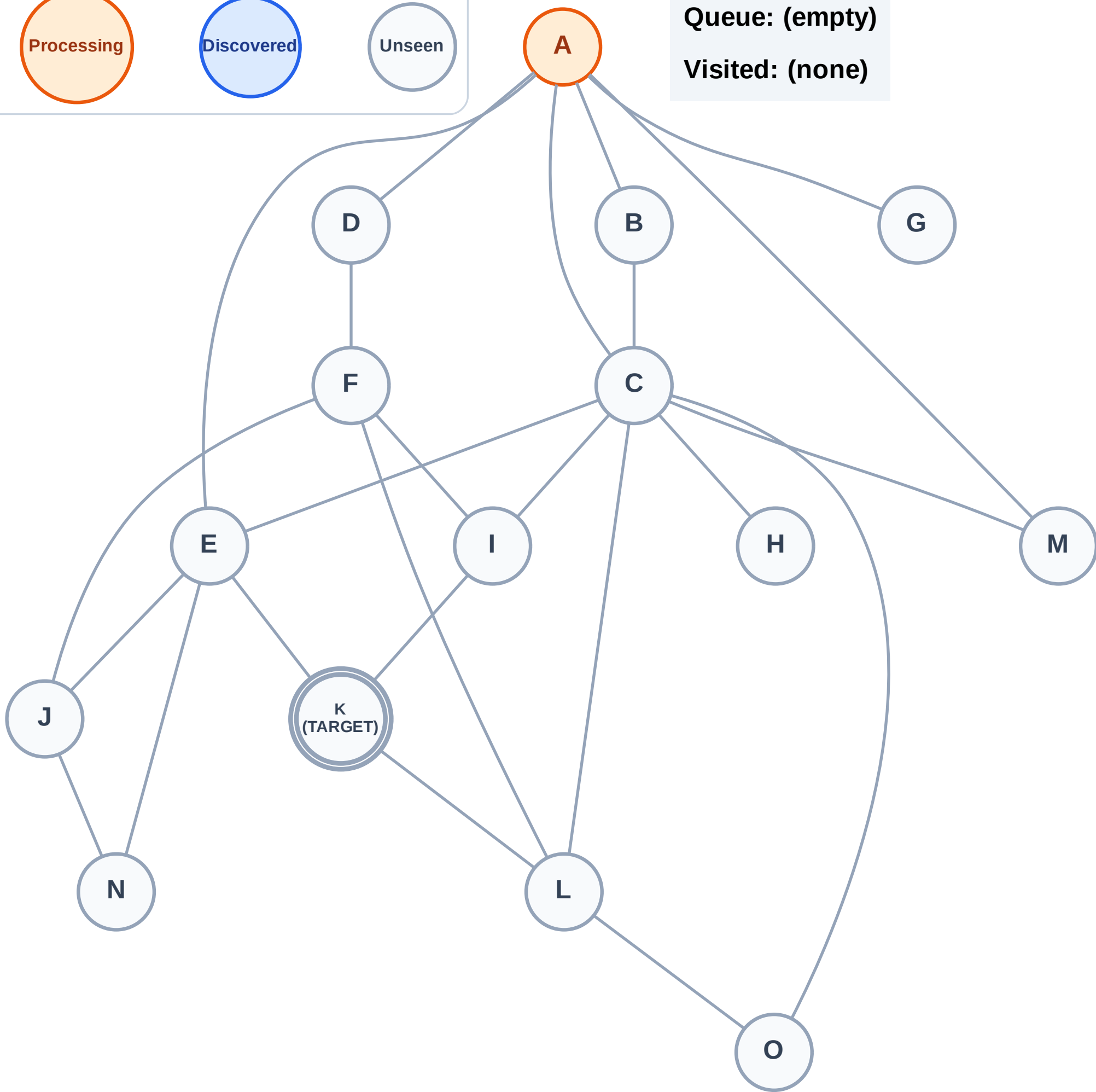
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)

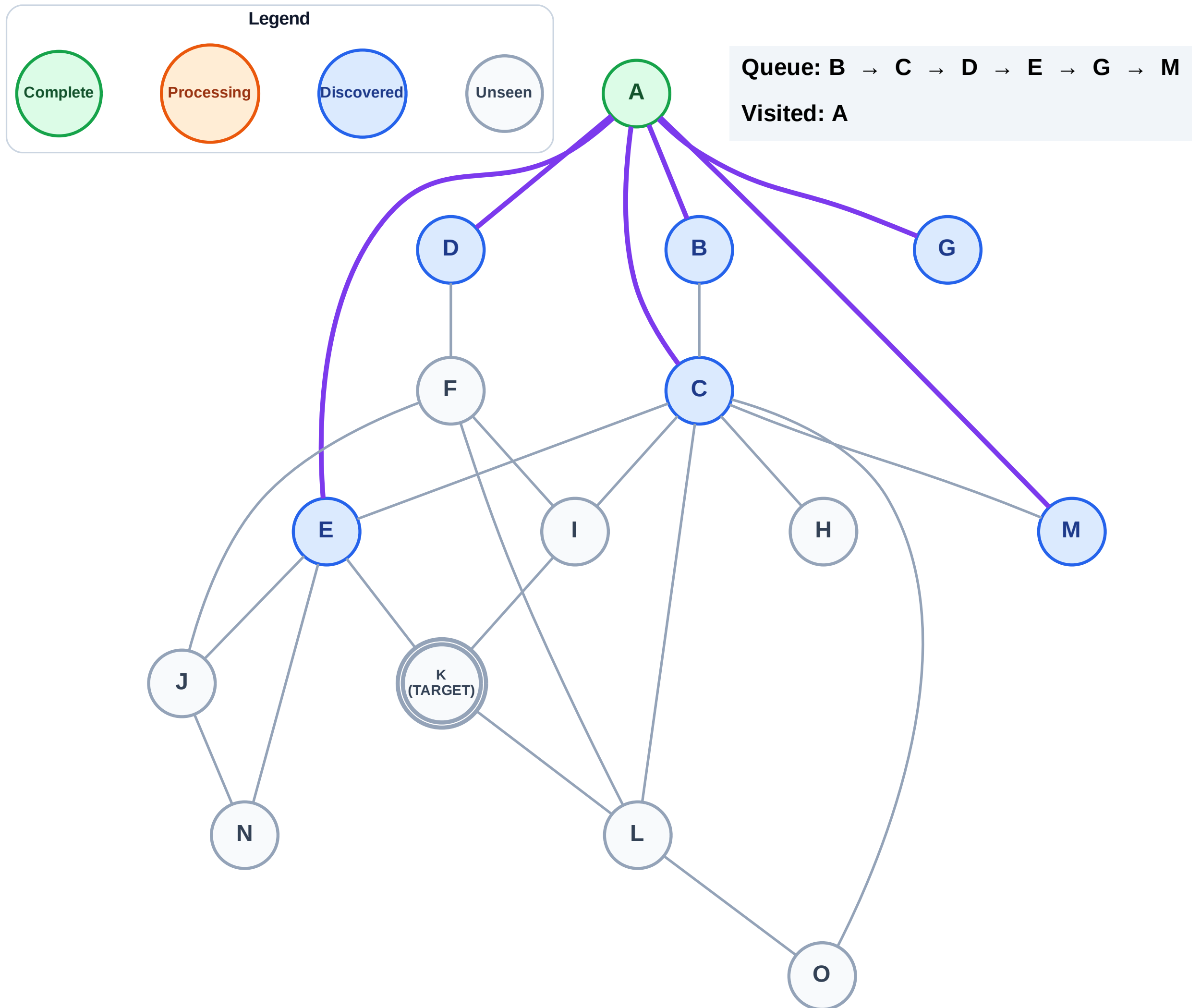


BFS Traversal

Step 3: Discover B, C, D, E, G, M from A

BFS Traversal

Step 3: Discover B, C, D, E, G, M from A



BFS Traversal

Step 4: Process node B

Legend

Complete

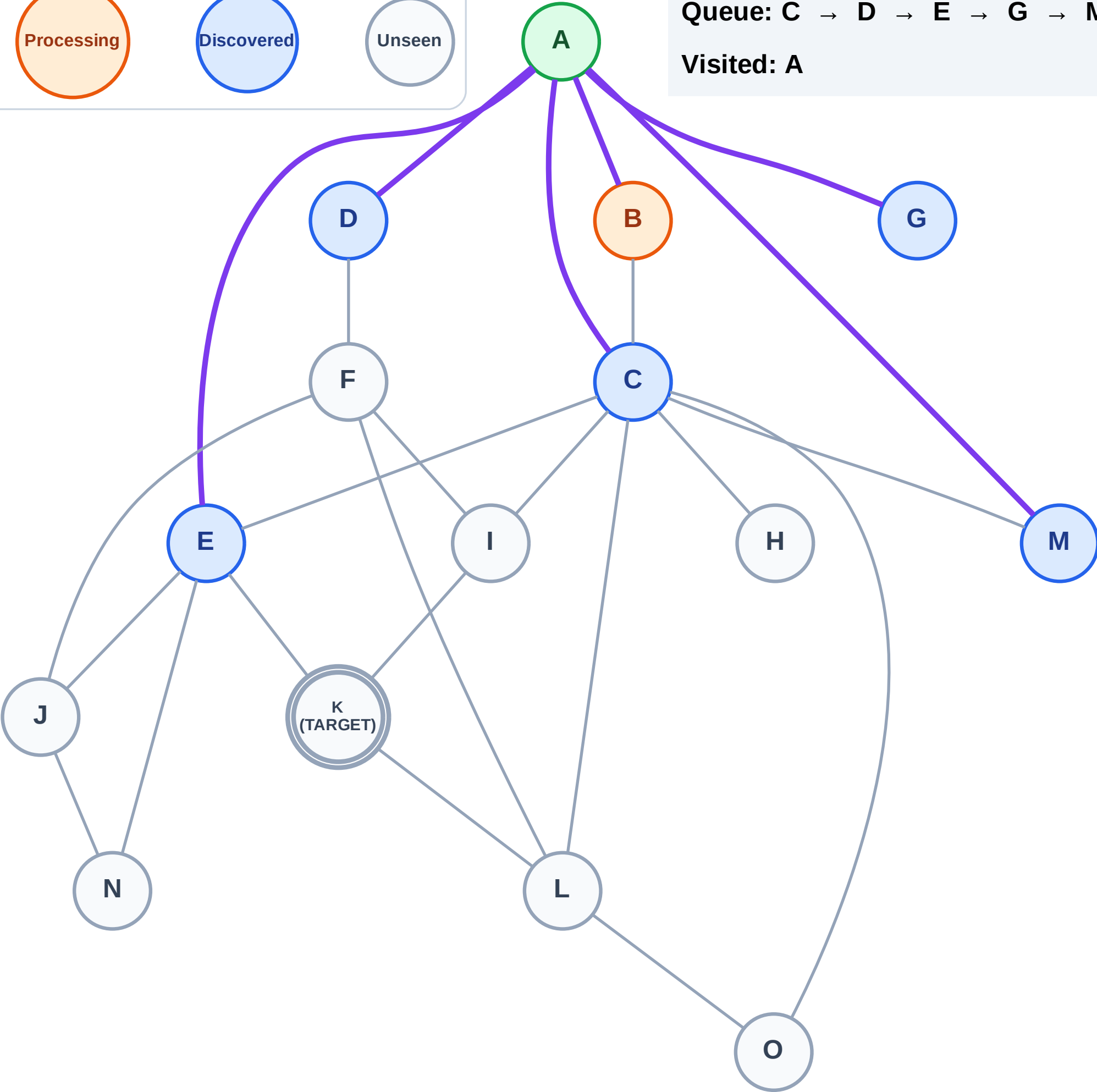
Processing

Discovered

Unseen

Queue: C → D → E → G → M

Visited: A



BFS Traversal

Step 5: No new neighbors from B

Legend

Complete

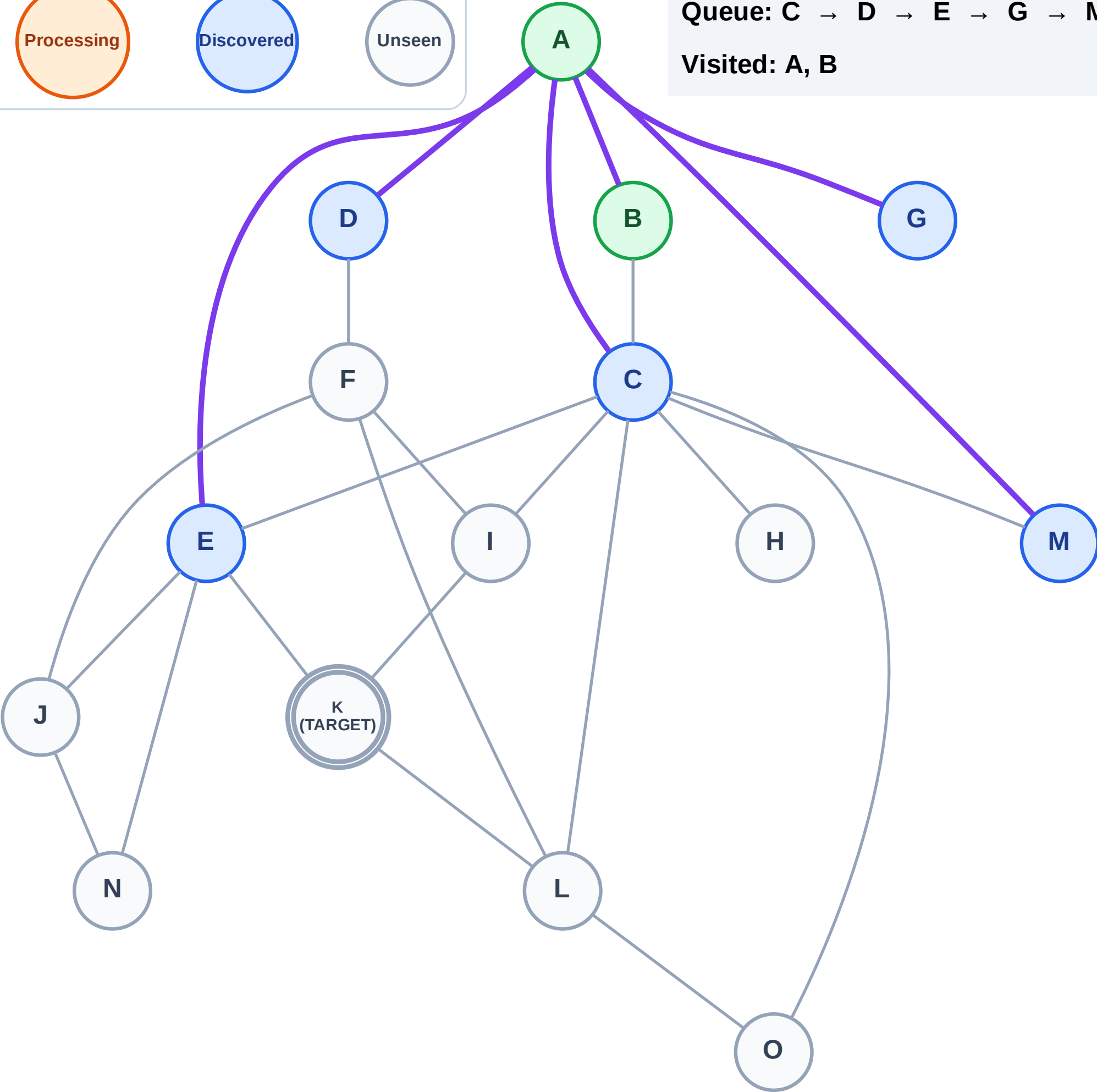
Processing

Discovered

Unseen

Queue: C → D → E → G → M

Visited: A, B



BFS Traversal

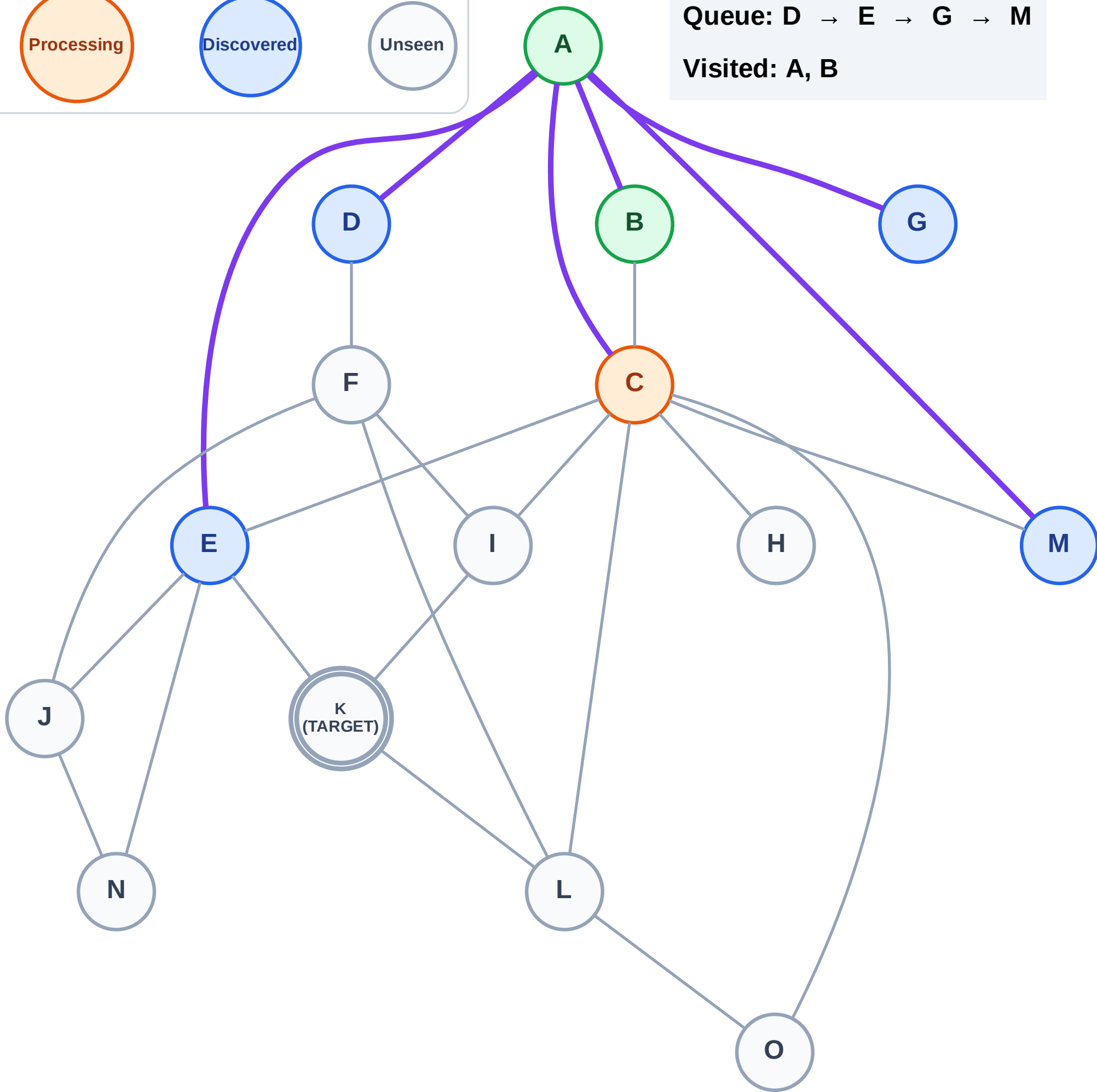
Step 6: Process node C

Legend



Queue: D → E → G → M

Visited: A, B



BFS Traversal

Step 7: Discover H, I, L, O from C

Legend

Complete

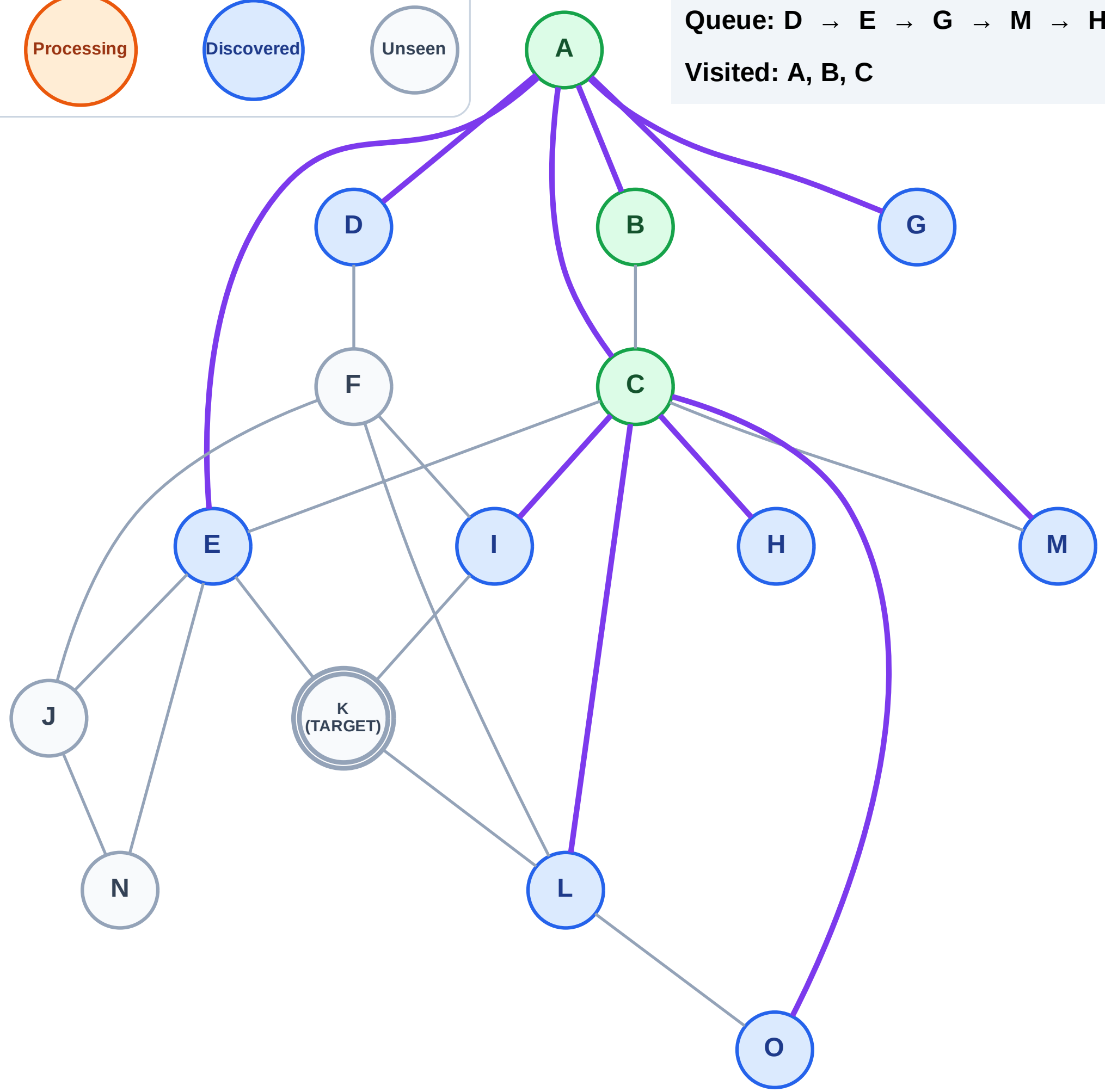
Processing

Discovered

Unseen

Queue: D → E → G → M → H → I → L → O

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

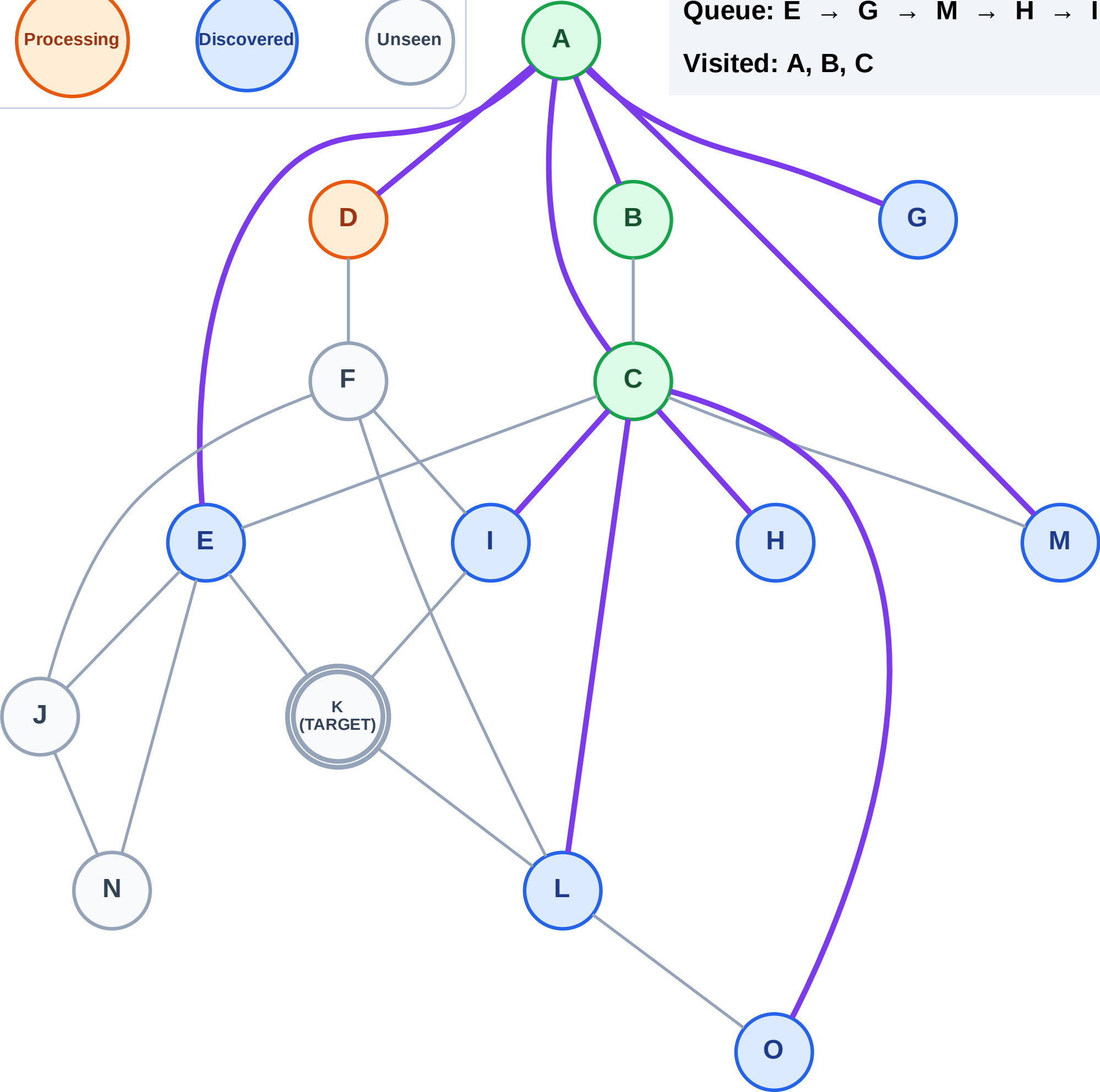
Processing

Discovered

Unseen

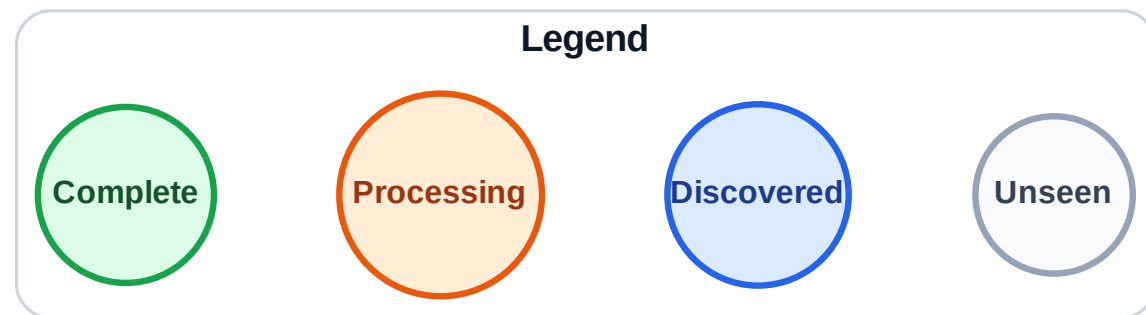
Queue: E → G → M → H → I → L → O

Visited: A, B, C



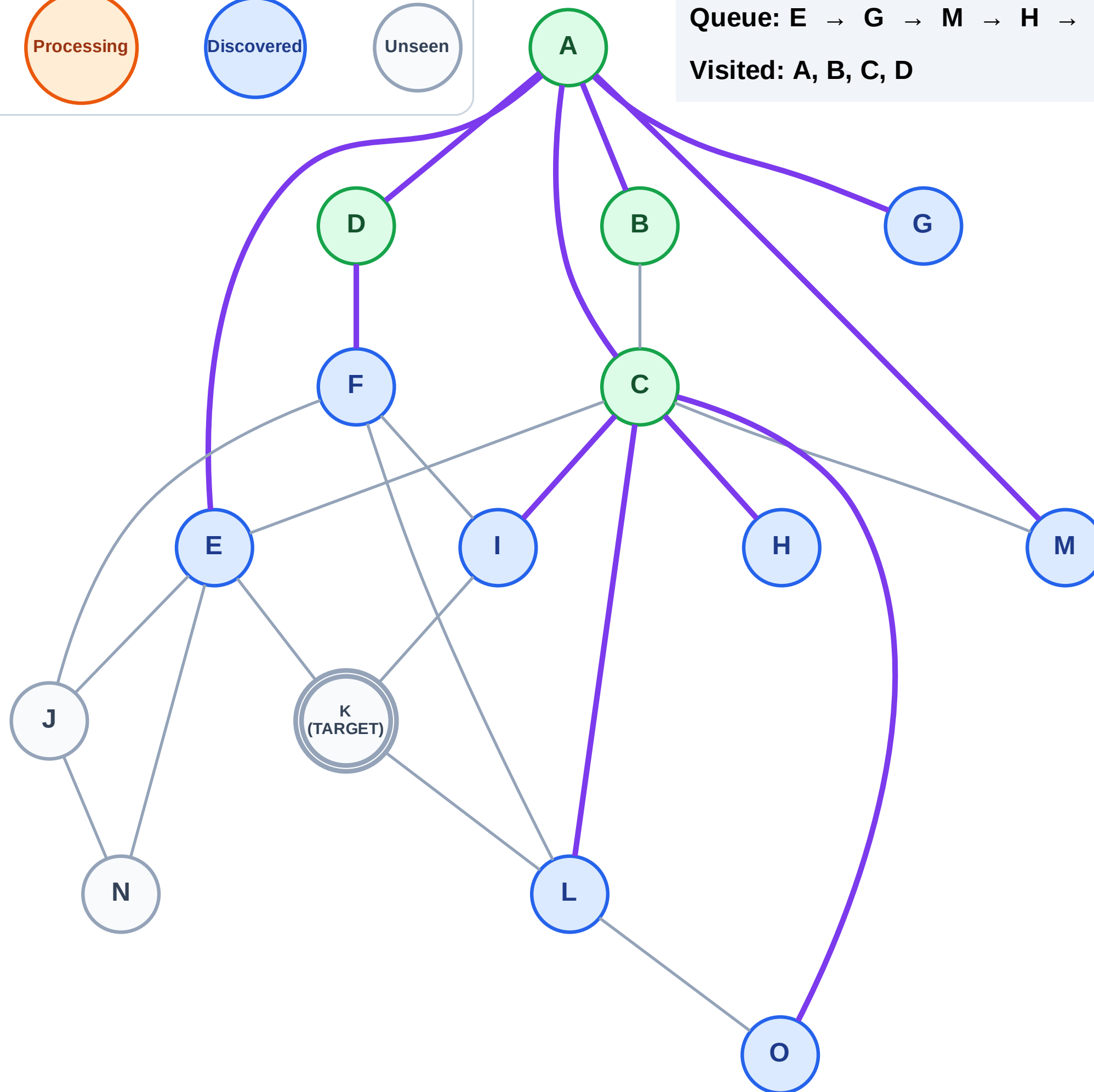
BFS Traversal

Step 9: Discover F from D



Queue: E → G → M → H → I → L → O → F

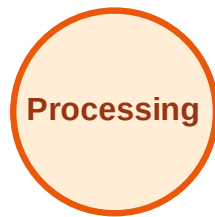
Visited: A, B, C, D



BFS Traversal

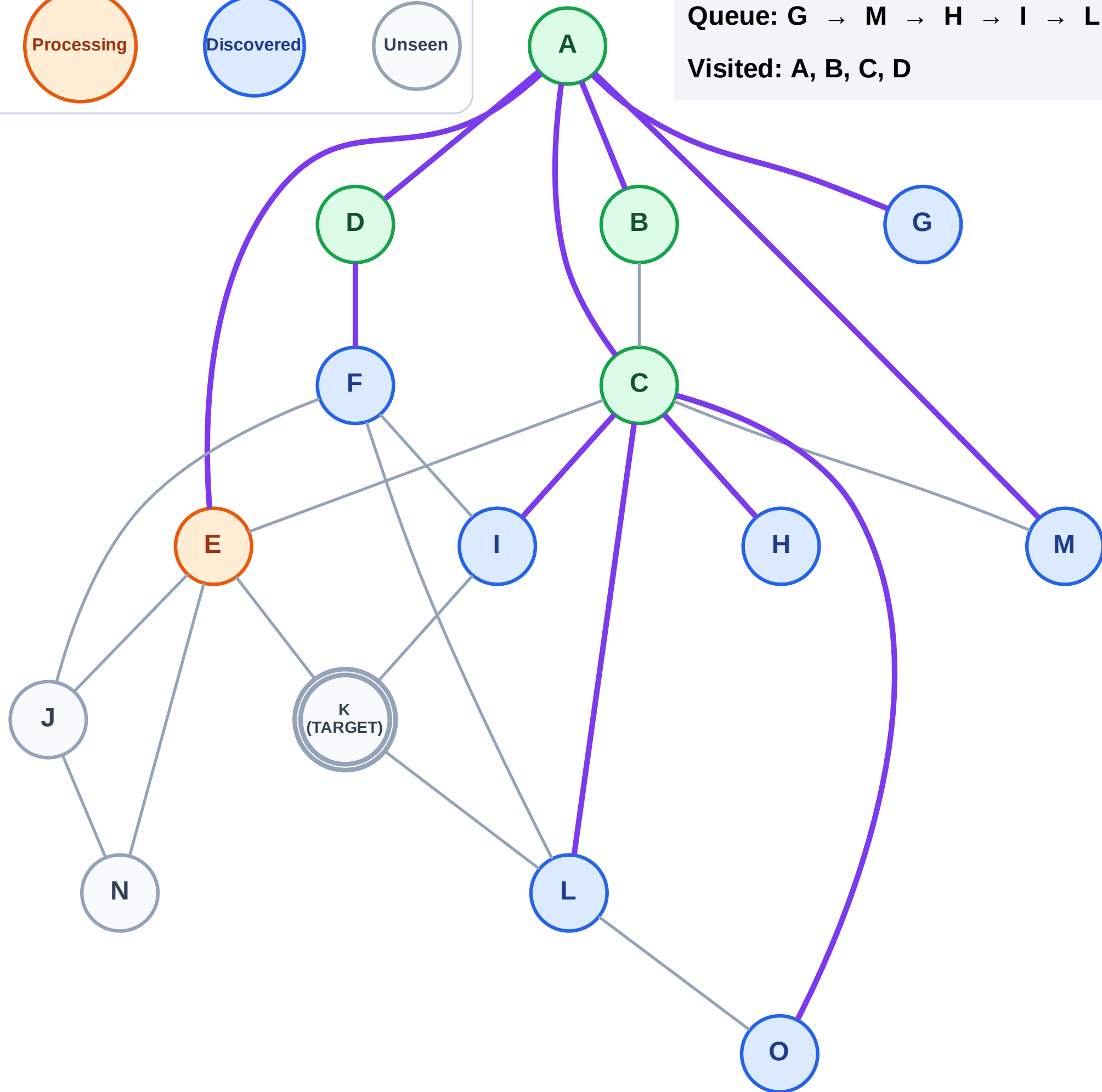
Step 10: Process node E

Legend



Queue: G → M → H → I → L → O → F

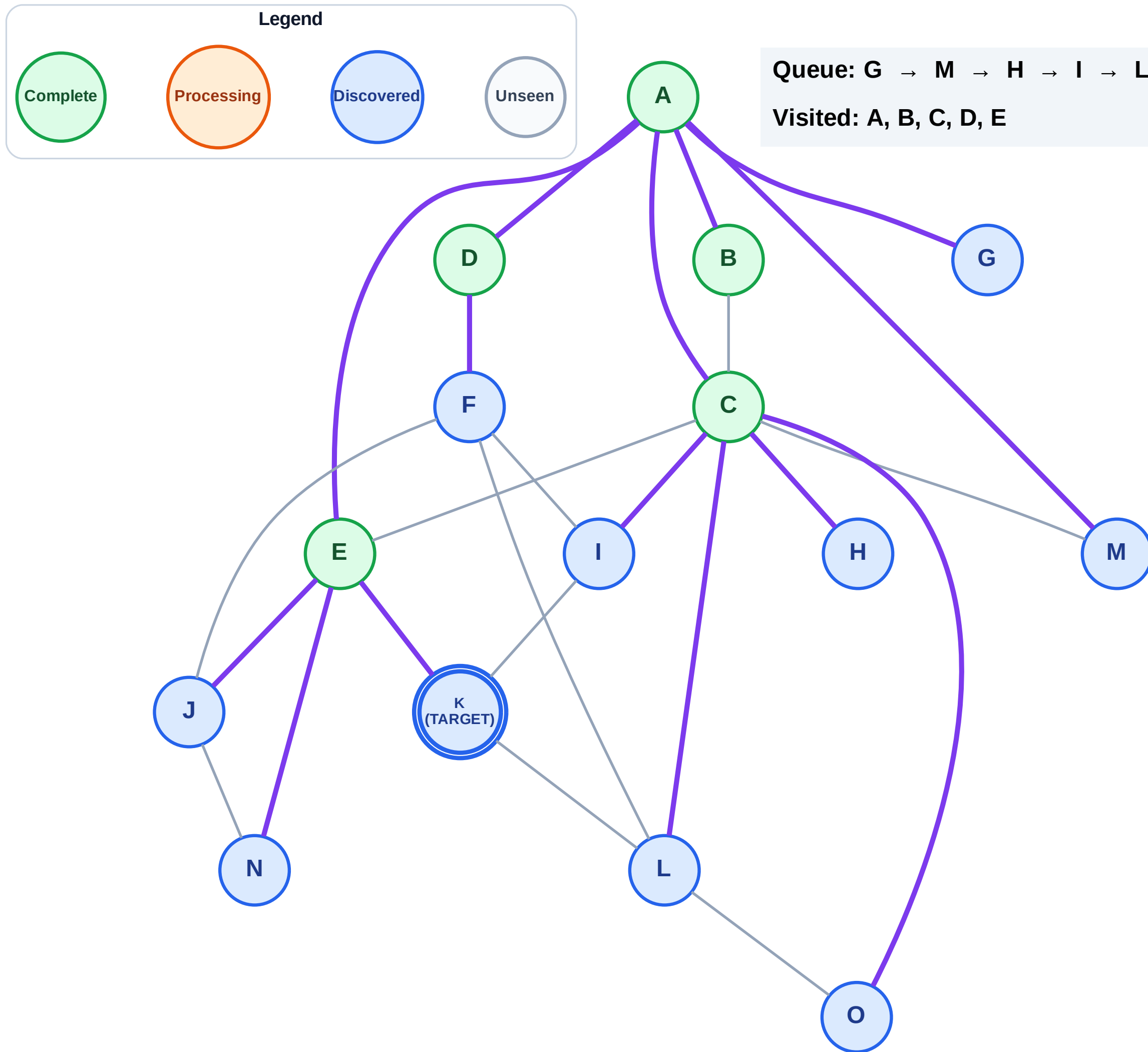
Visited: A, B, C, D



Step 11: Discover J, K, N from E

Queue: G → M → H → I → L → O → F → J → K → N

Visited: A, B, C, D, E



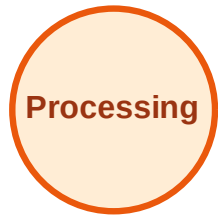
BFS Traversal

Step 12: Process node G

Legend



Complete



Processing



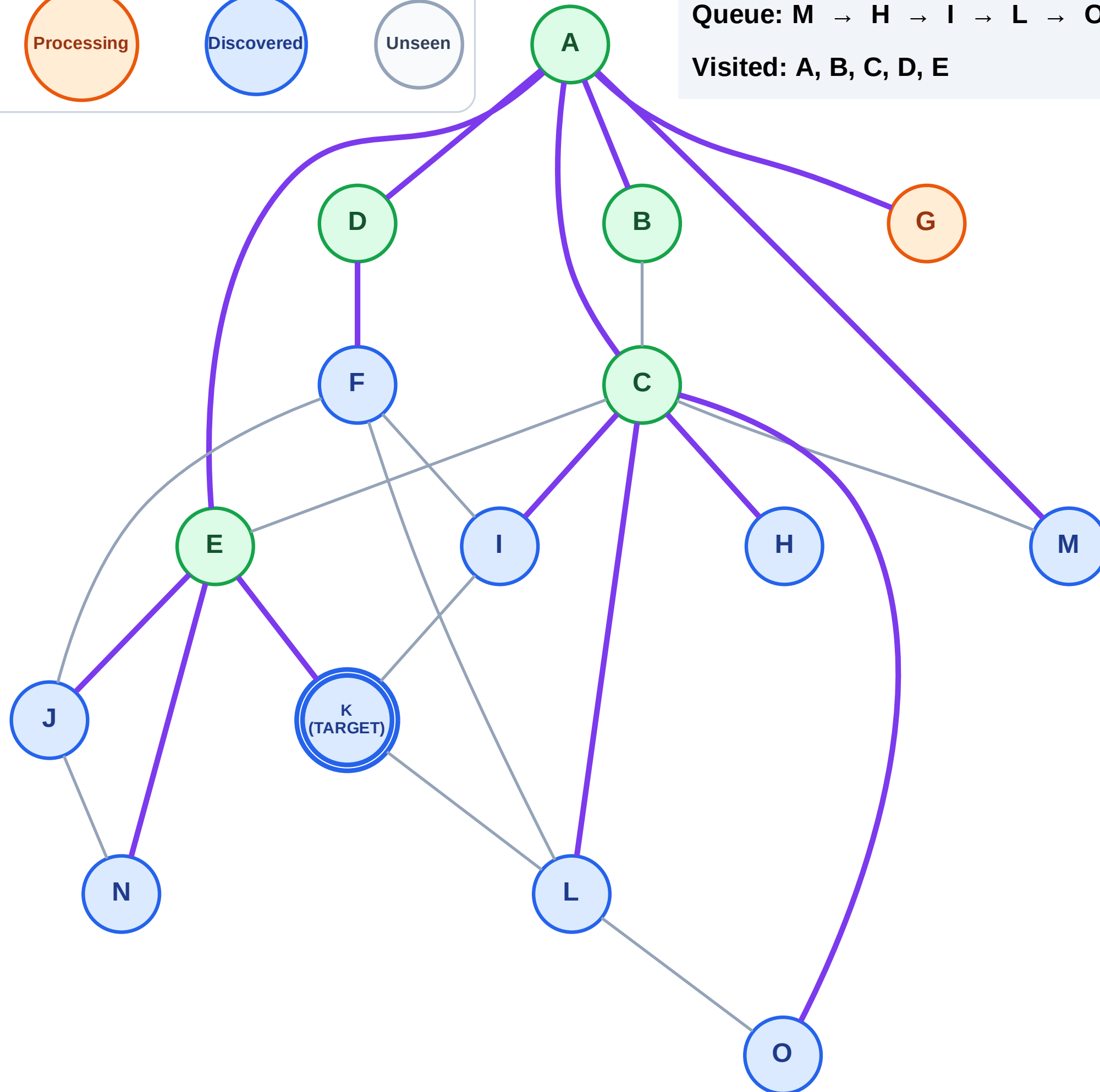
Discovered



Unseen

Queue: M → H → I → L → O → F → J → K → N

Visited: A, B, C, D, E



BFS Traversal

Step 13: No new neighbors from G

Legend

Complete

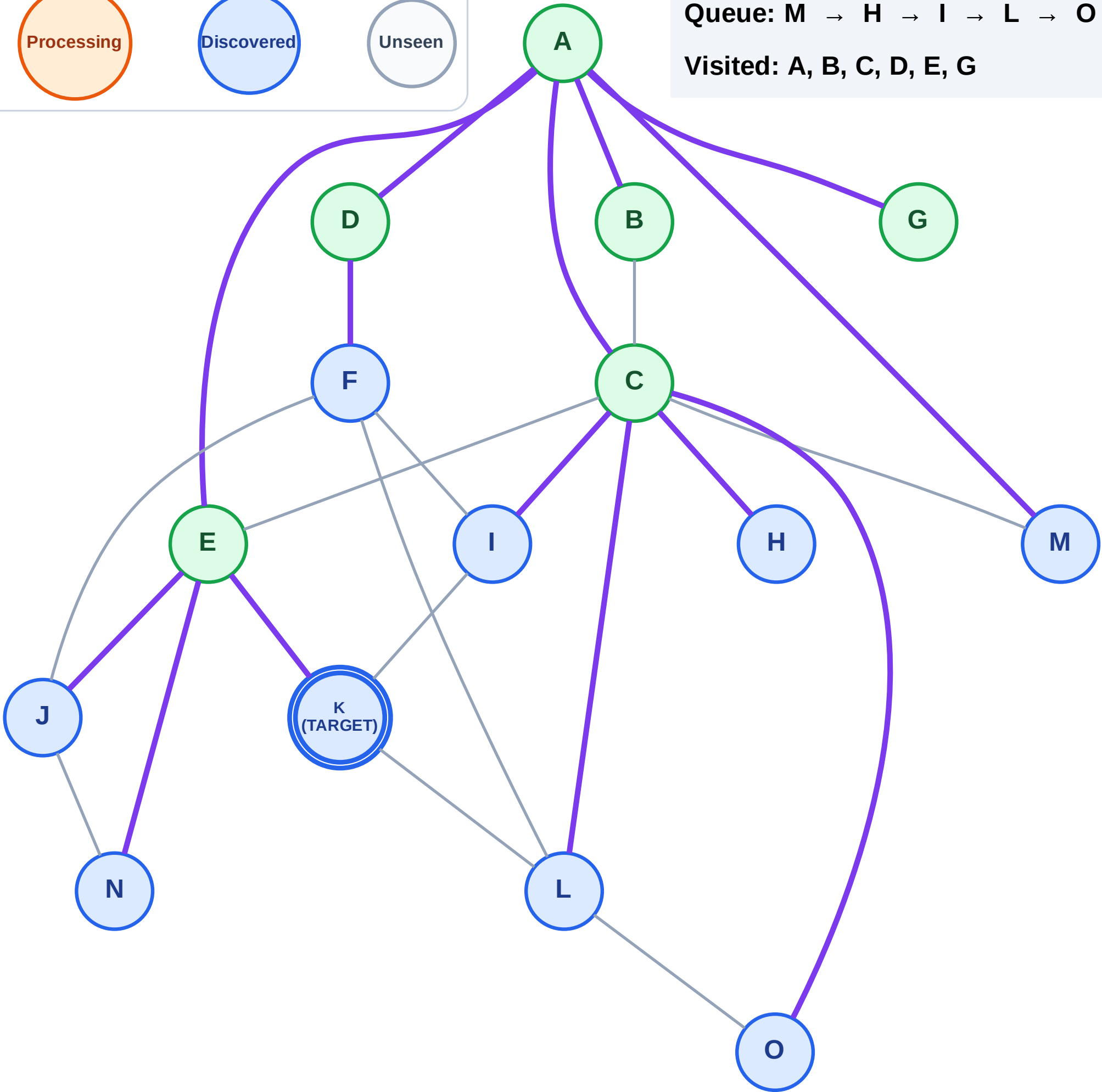
Processing

Discovered

Unseen

Queue: M → H → I → L → O → F → J → K → N

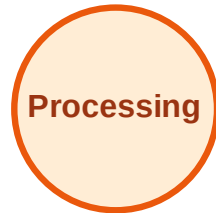
Visited: A, B, C, D, E, G



BFS Traversal

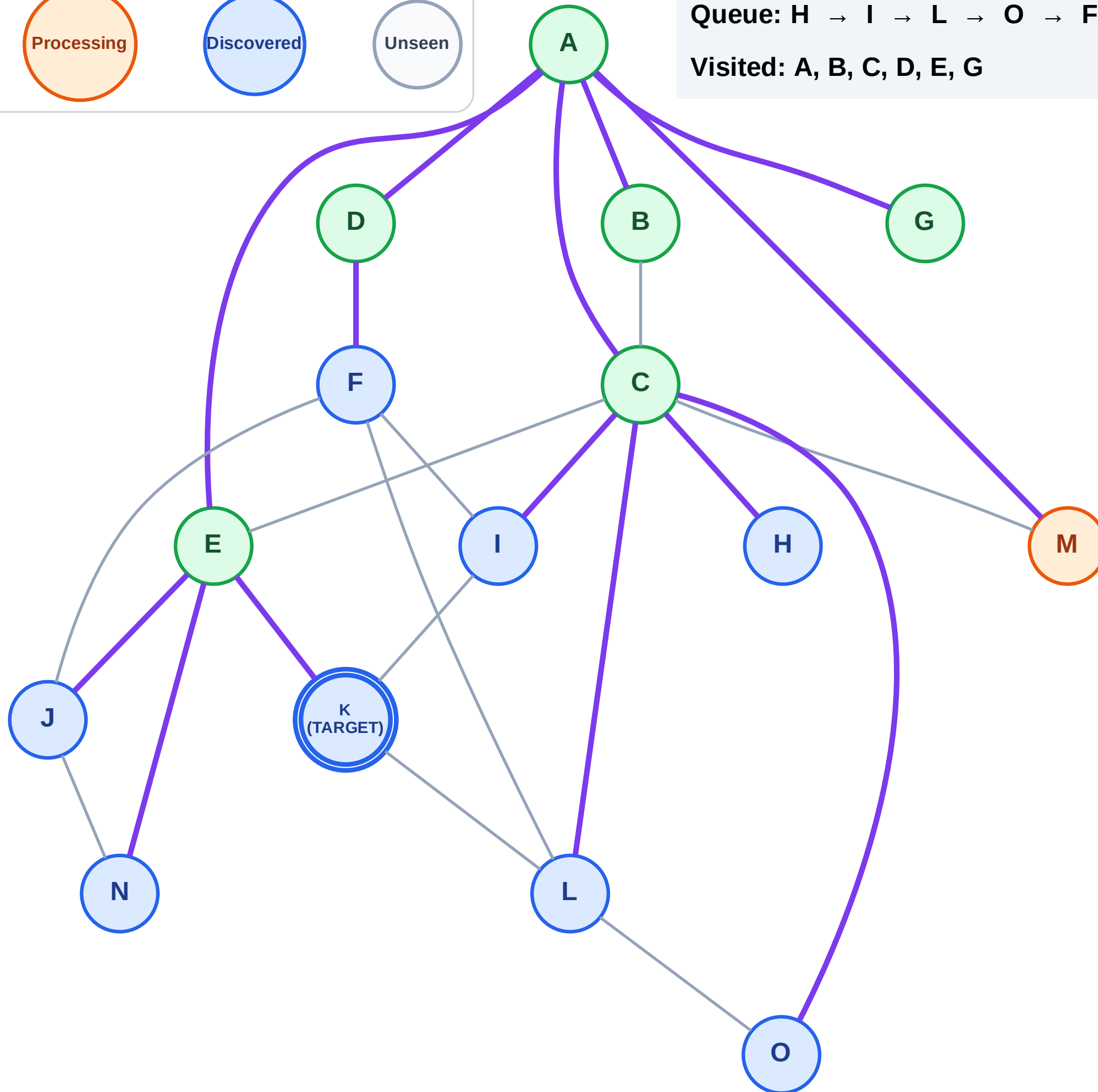
Step 14: Process node M

Legend



Queue: H → I → L → O → F → J → K → N

Visited: A, B, C, D, E, G



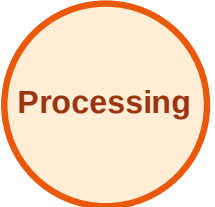
BFS Traversal

Step 15: No new neighbors from M

Legend



Complete



Processing



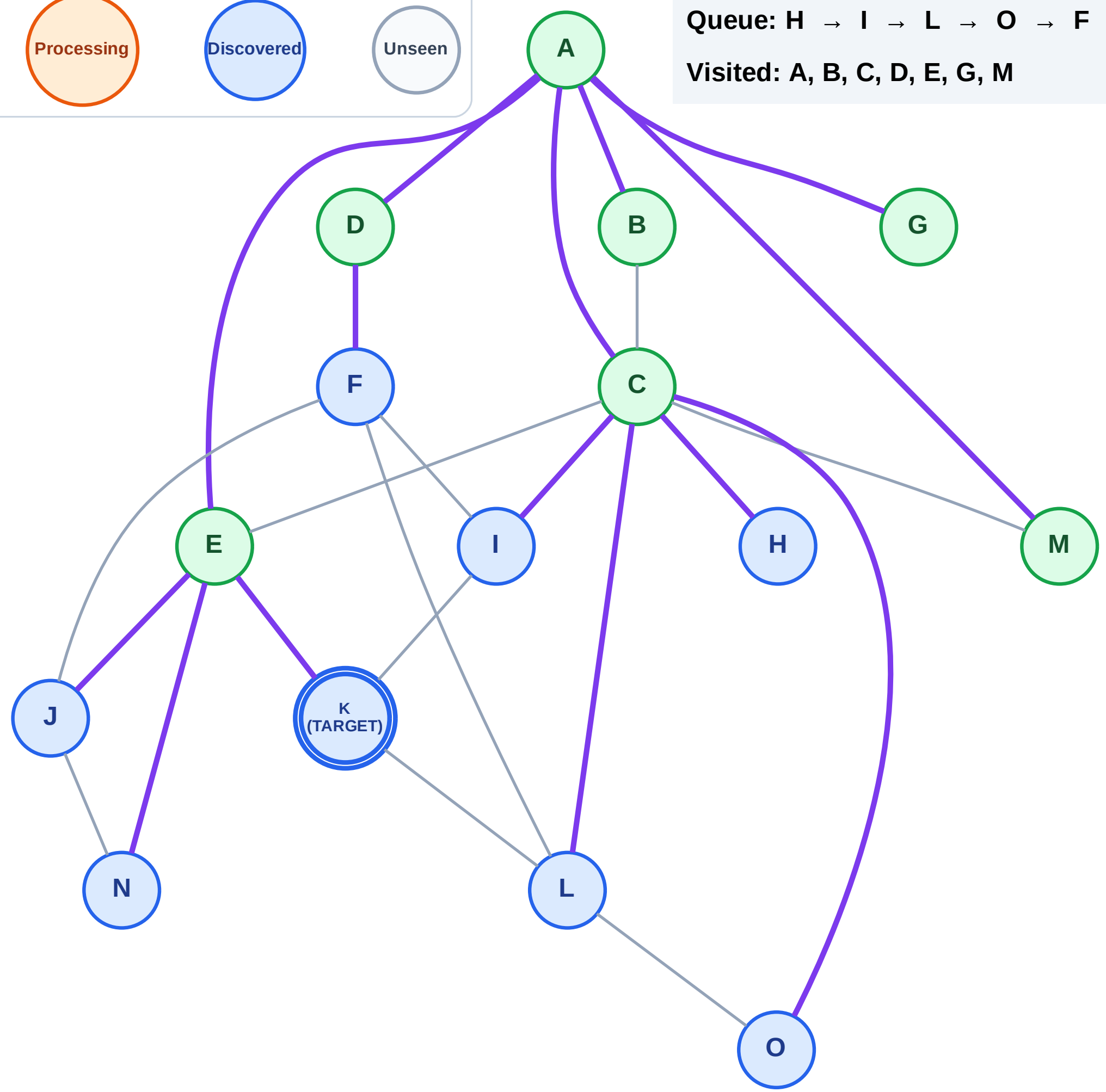
Discovered



Unseen

Queue: H → I → L → O → F → J → K → N

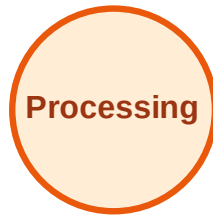
Visited: A, B, C, D, E, G, M



BFS Traversal

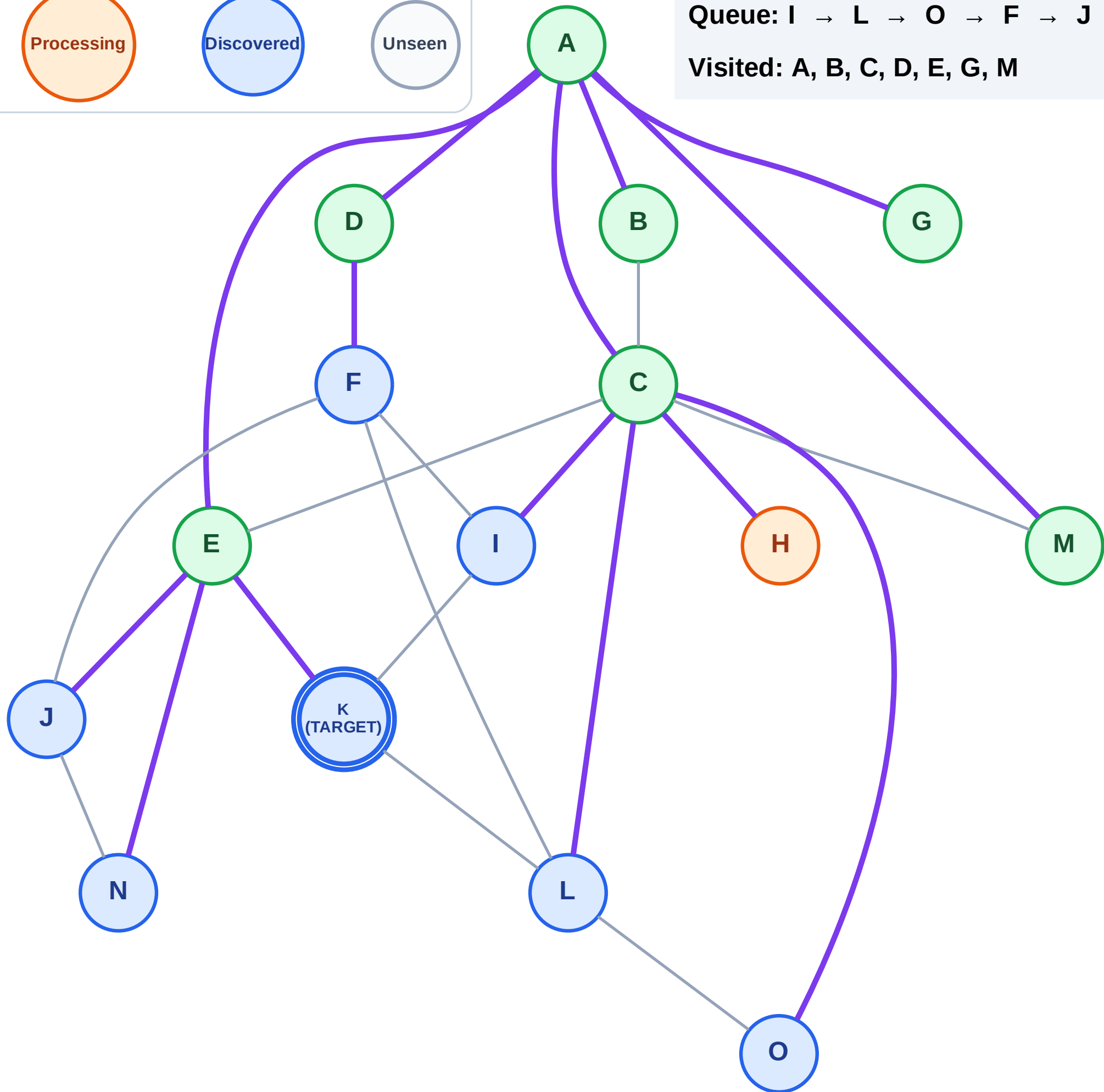
Step 16: Process node H

Legend



Queue: I → L → O → F → J → K → N

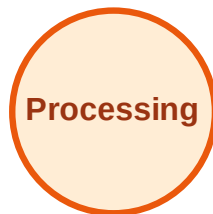
Visited: A, B, C, D, E, G, M



BFS Traversal

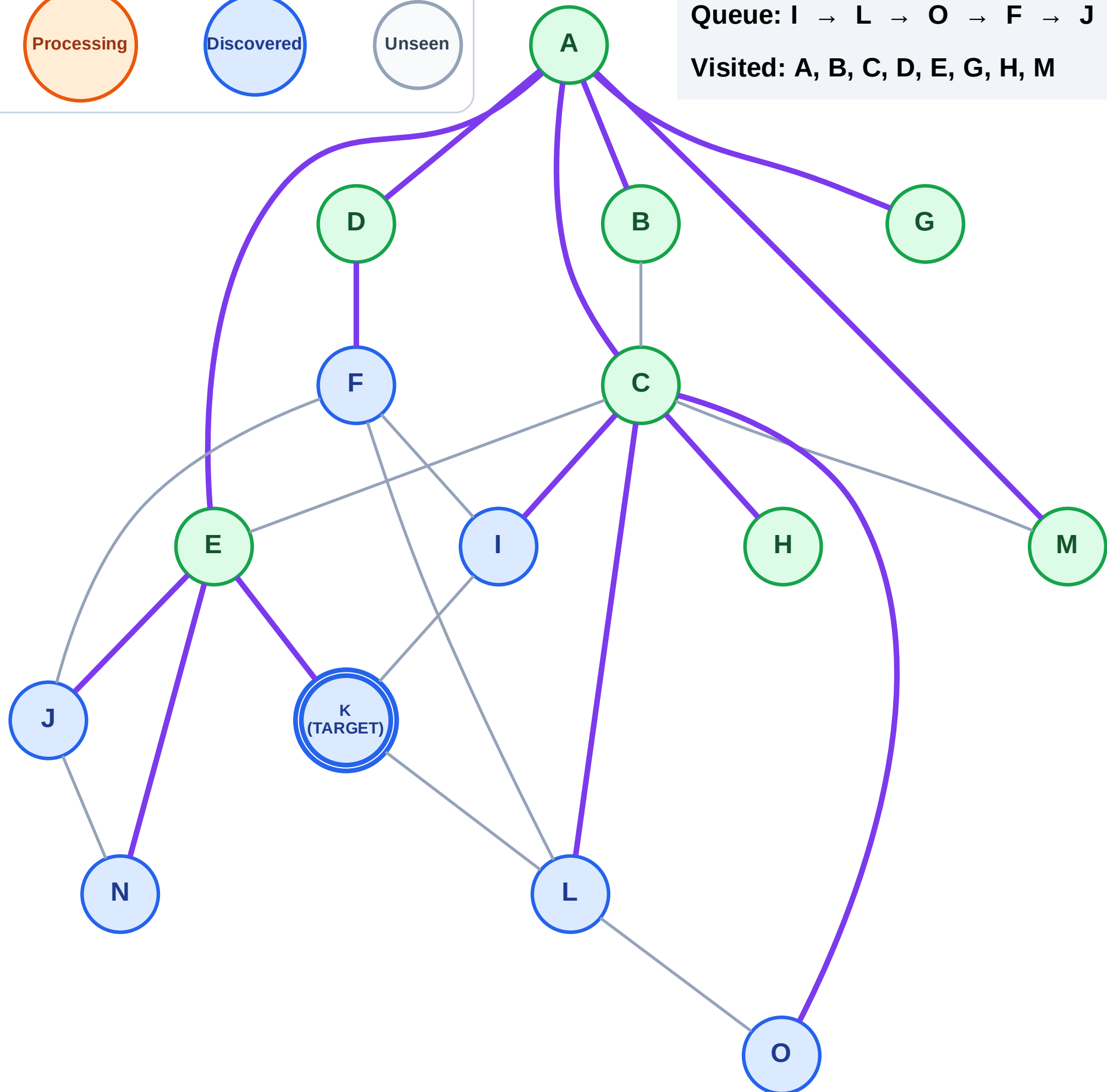
Step 17: No new neighbors from H

Legend



Queue: I → L → O → F → J → K → N

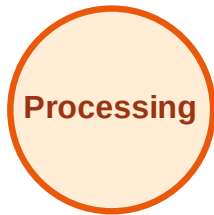
Visited: A, B, C, D, E, G, H, M



BFS Traversal

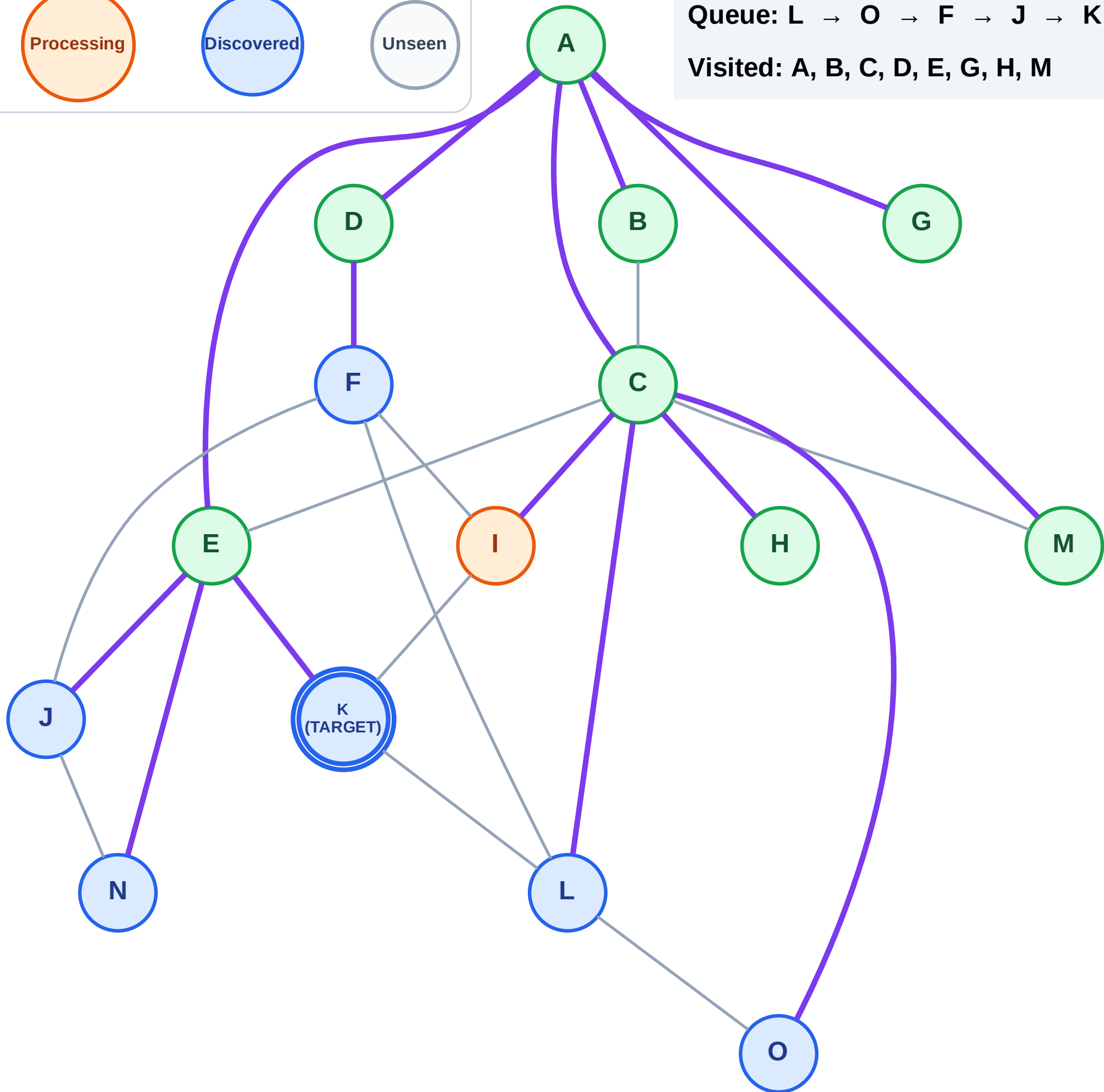
Step 18: Process node I

Legend



Queue: L → O → F → J → K → N

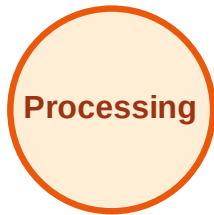
Visited: A, B, C, D, E, G, H, M



BFS Traversal

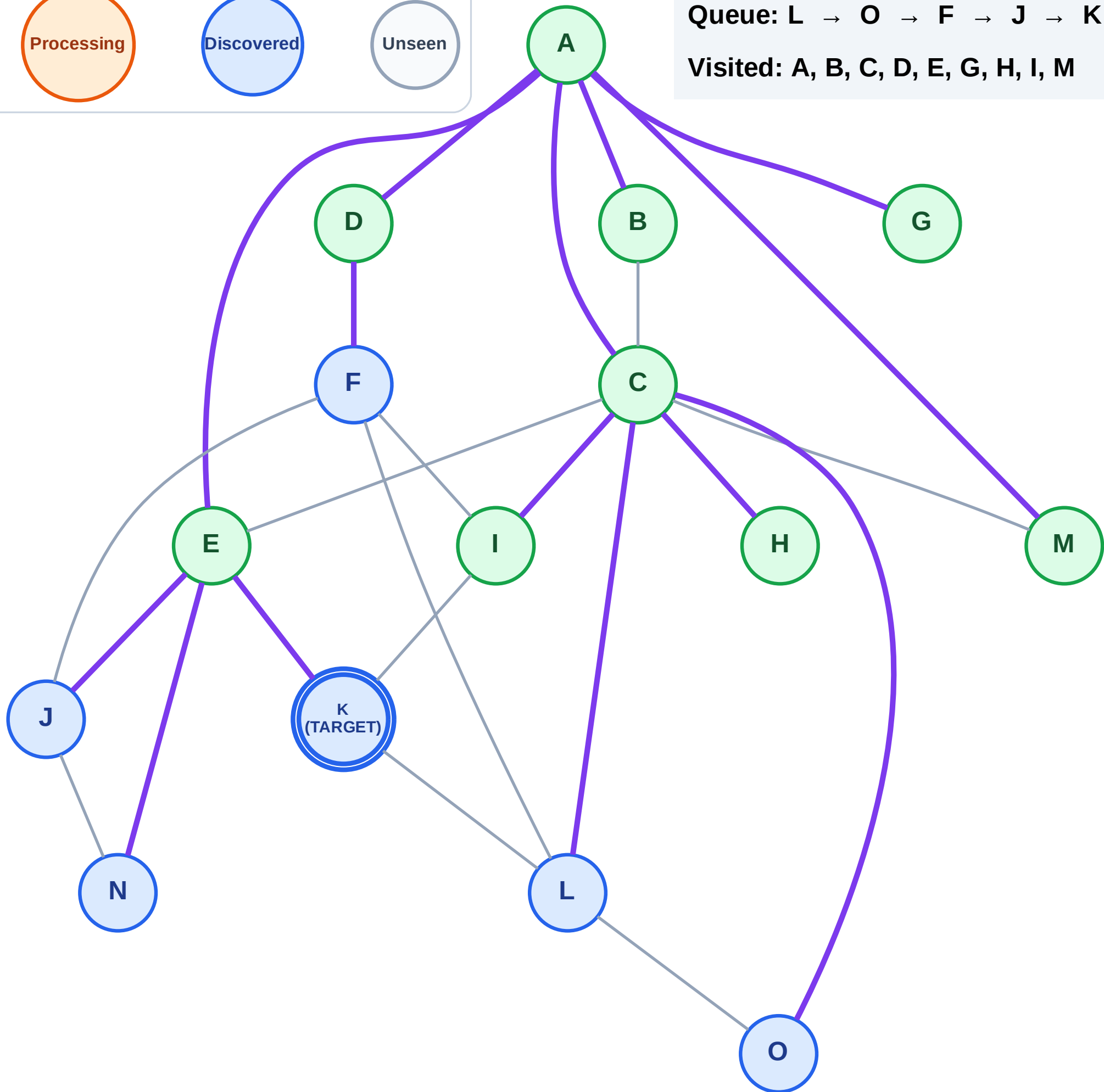
Step 19: No new neighbors from I

Legend



Queue: L → O → F → J → K → N

Visited: A, B, C, D, E, G, H, I, M



BFS Traversal

Step 20: Process node L

Legend



Complete



Processing



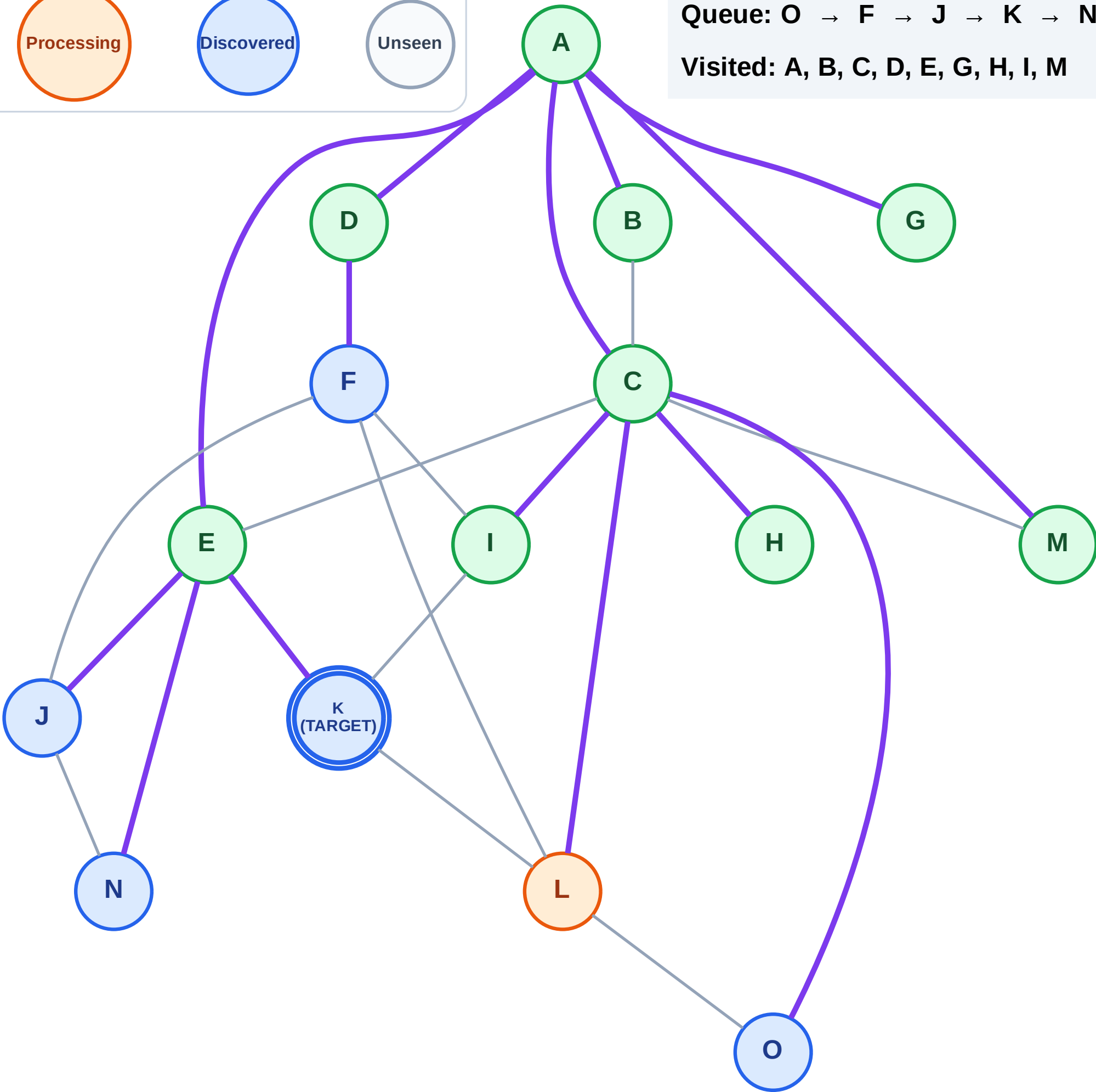
Discovered



Unseen

Queue: O → F → J → K → N

Visited: A, B, C, D, E, G, H, I, M



BFS Traversal

Step 21: No new neighbors from L

Legend

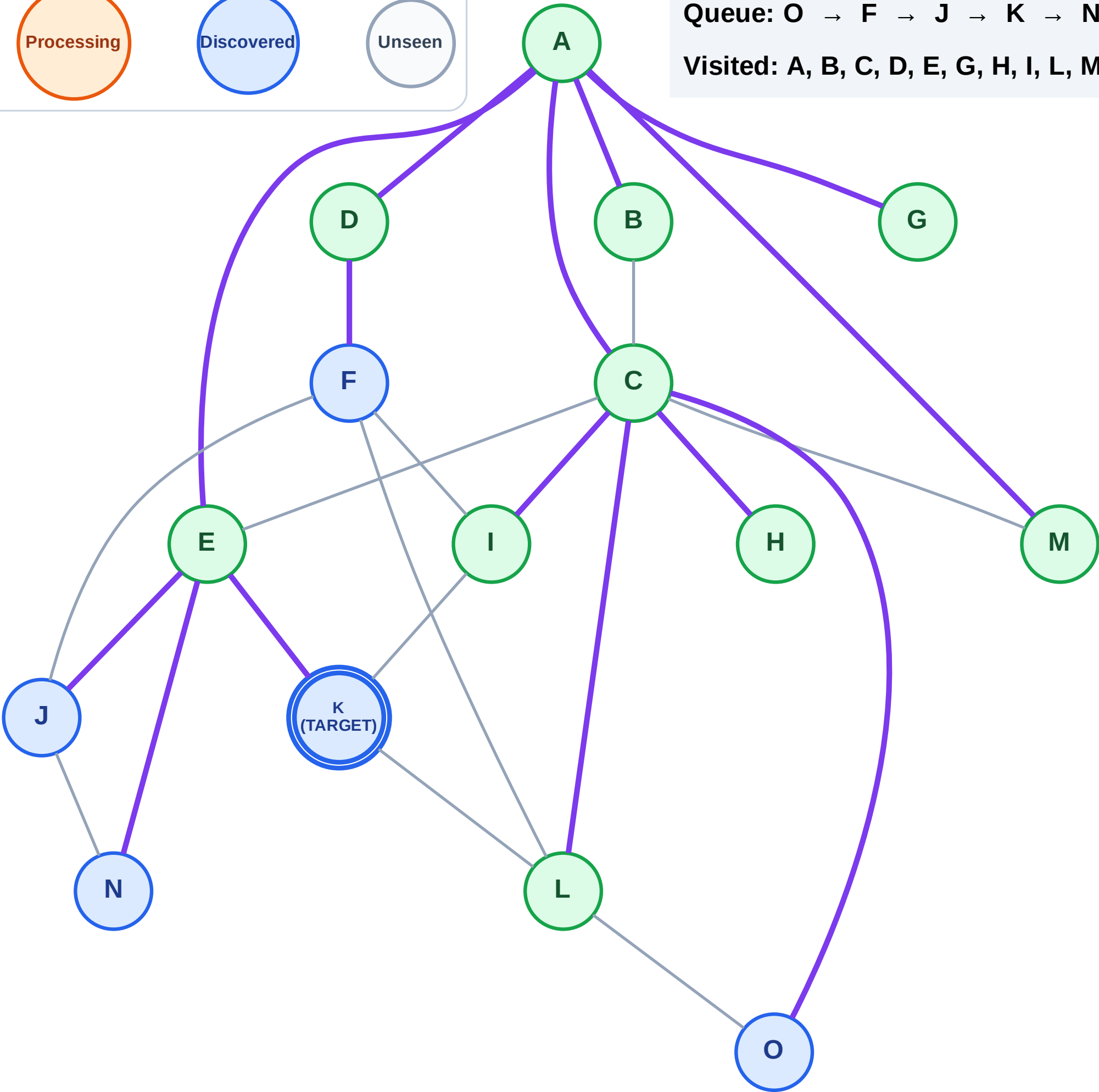
Complete

Processing

Discovered

Unseen

Queue: O → F → J → K → N
Visited: A, B, C, D, E, G, H, I, L, M



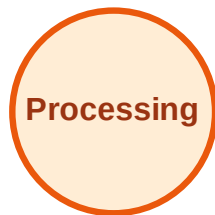
BFS Traversal

Step 22: Process node O

Legend



Complete



Processing



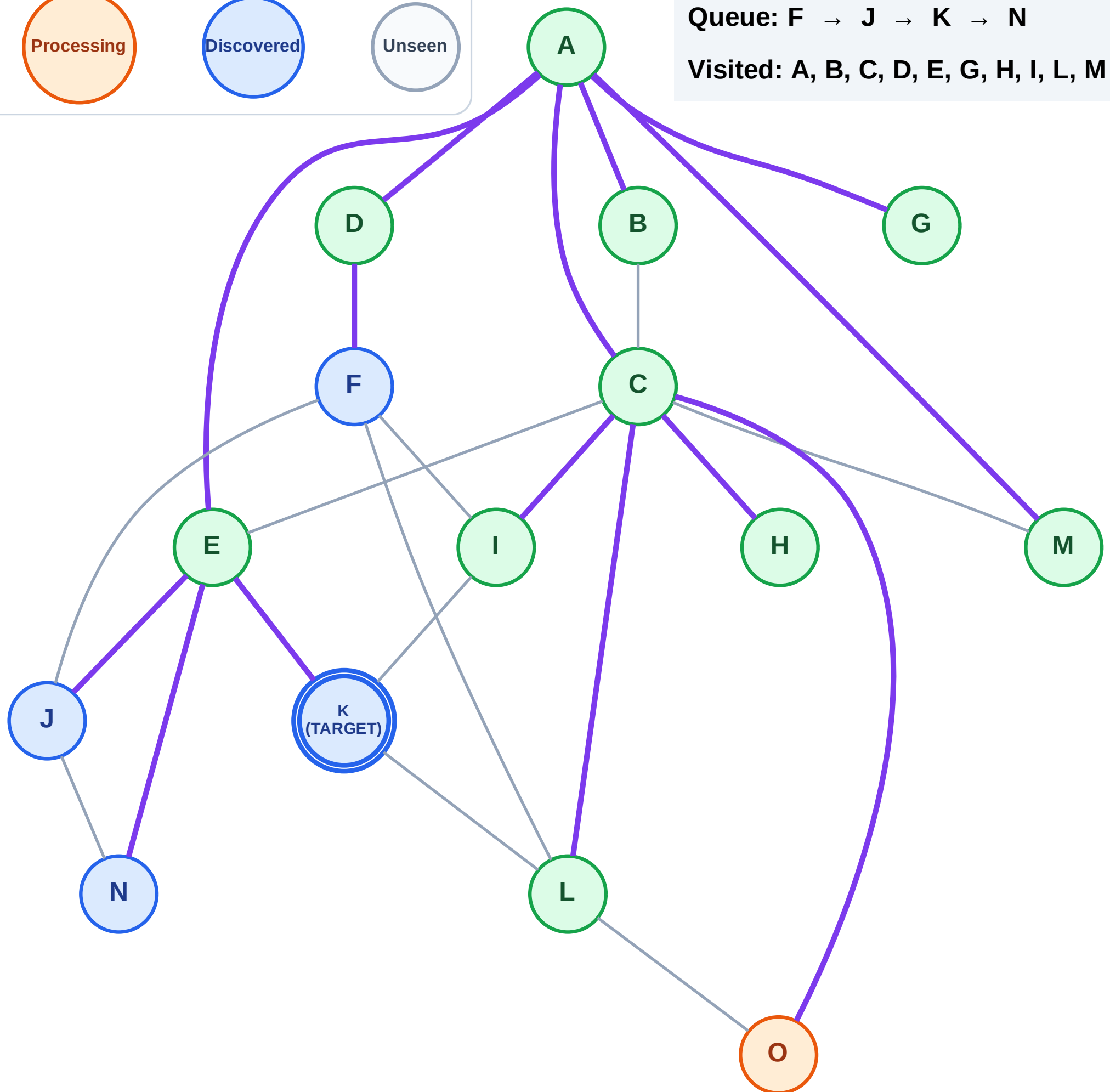
Discovered



Unseen

Queue: F → J → K → N

Visited: A, B, C, D, E, G, H, I, L, M



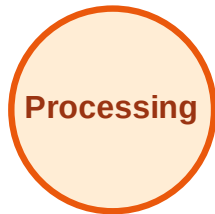
BFS Traversal

Step 23: No new neighbors from O

Legend



Complete



Processing



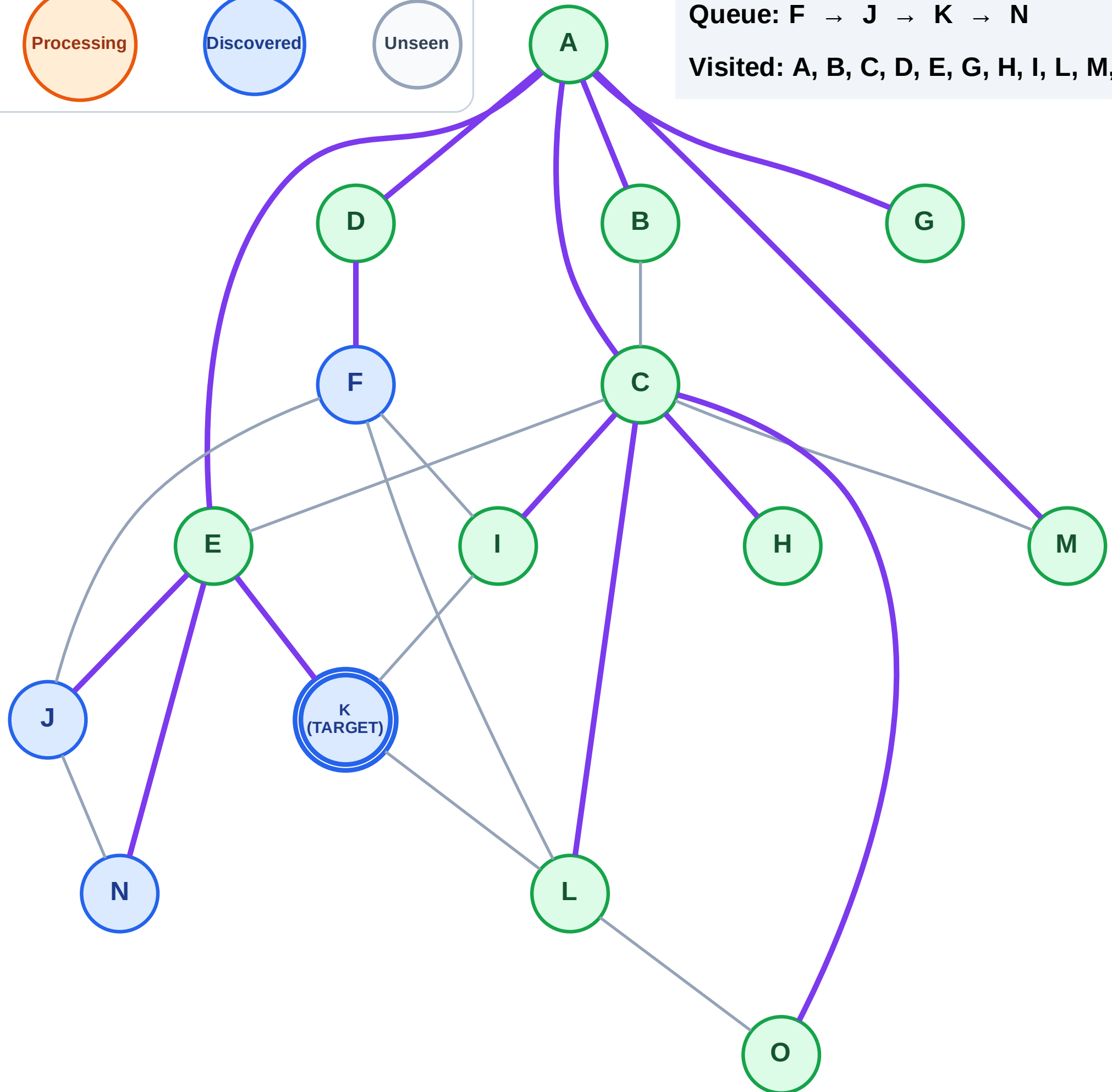
Discovered



Unseen

Queue: F → J → K → N

Visited: A, B, C, D, E, G, H, I, L, M, O



BFS Traversal

Step 24: Process node F

Legend

Complete

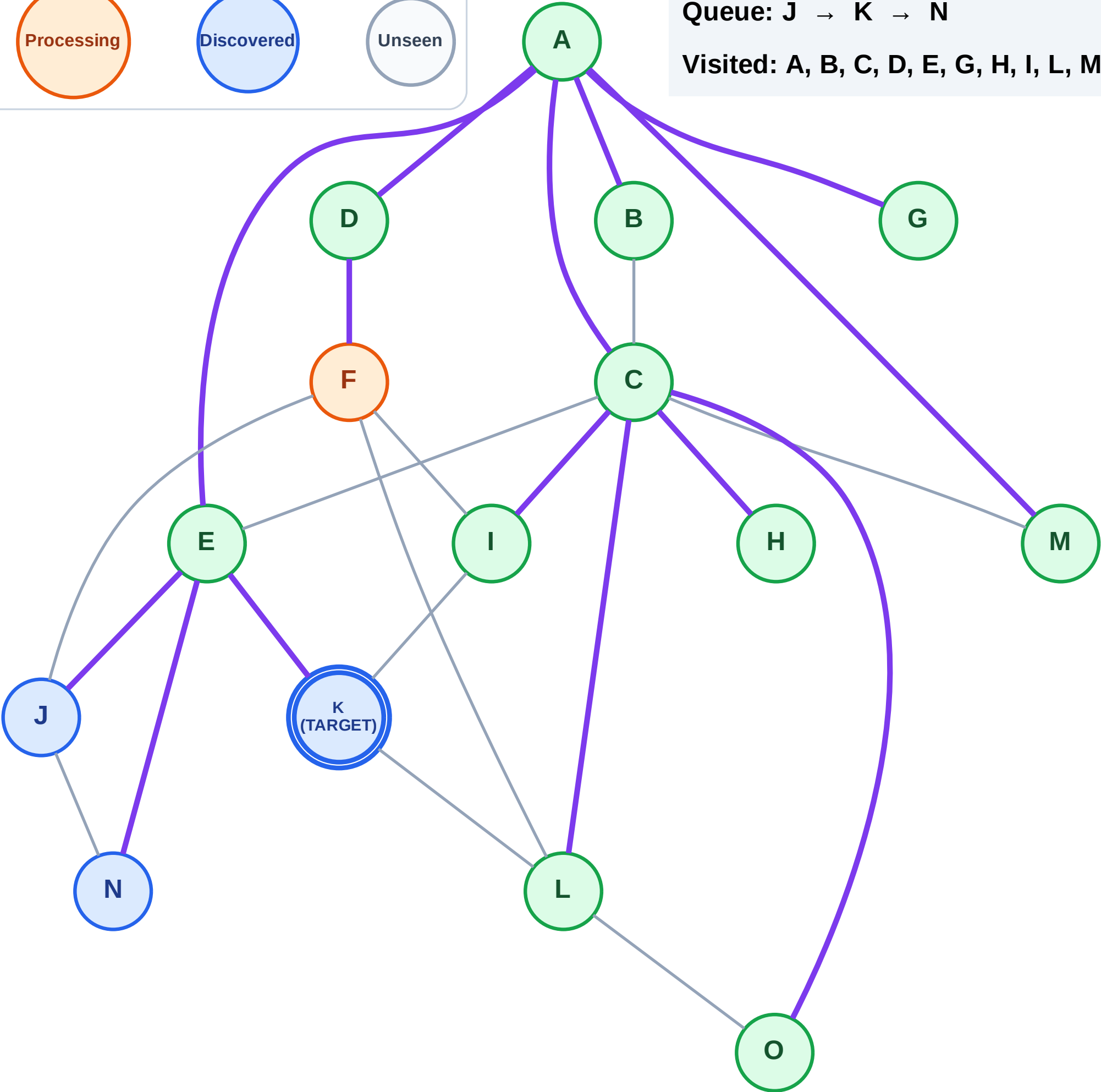
Processing

Discovered

Unseen

Queue: J → K → N

Visited: A, B, C, D, E, G, H, I, L, M, O



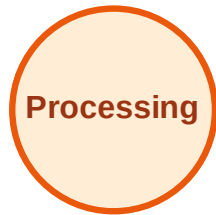
BFS Traversal

Step 25: No new neighbors from F

Legend



Complete



Processing



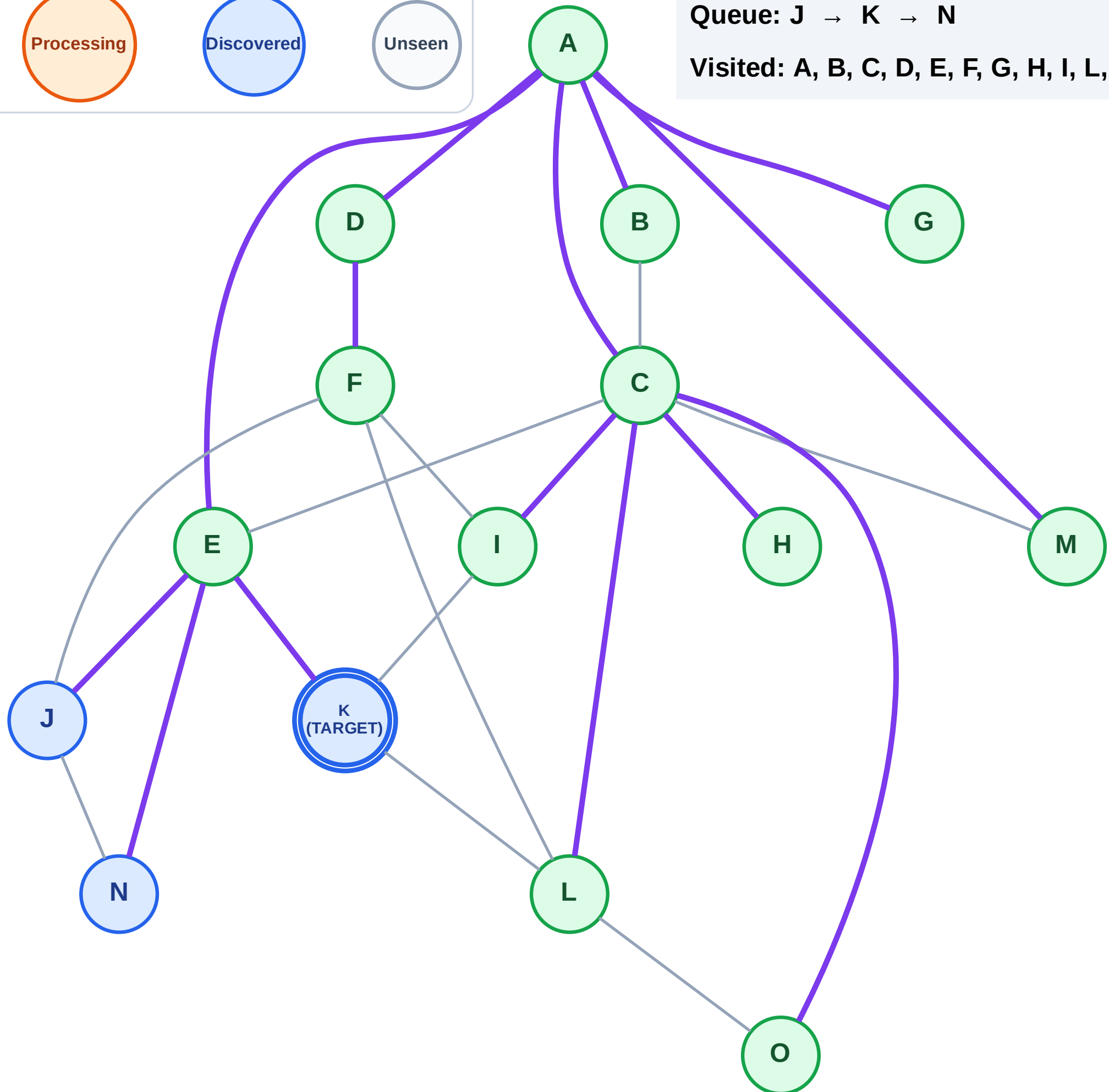
Discovered



Unseen

Queue: J → K → N

Visited: A, B, C, D, E, F, G, H, I, L, M, O



BFS Traversal

Step 26: Process node J

Legend

Complete

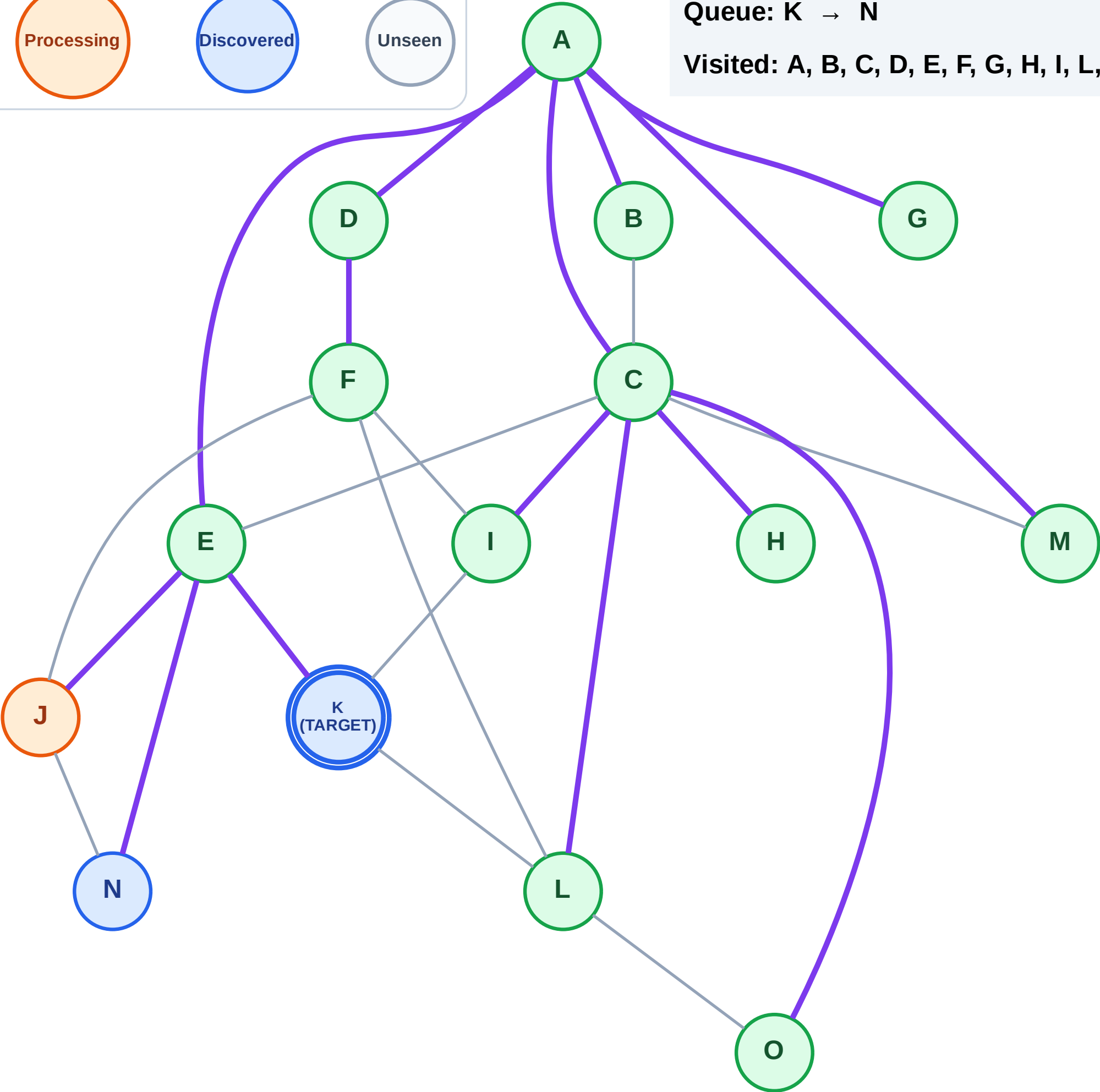
Processing

Discovered

Unseen

Queue: K → N

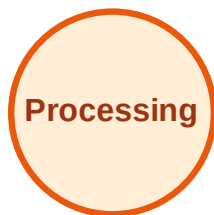
Visited: A, B, C, D, E, F, G, H, I, L, M, O



BFS Traversal

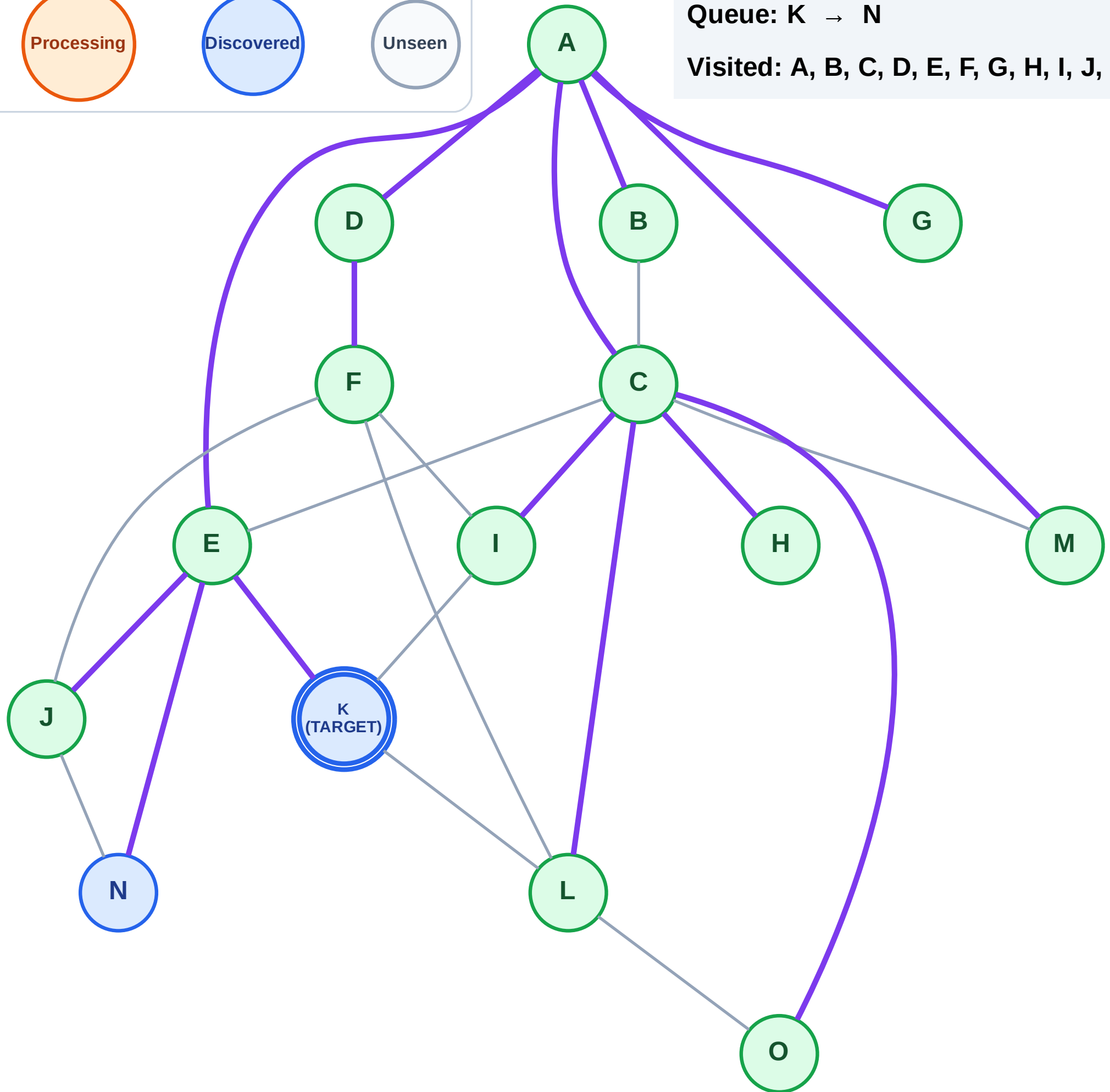
Step 27: No new neighbors from J

Legend



Queue: K → N

Visited: A, B, C, D, E, F, G, H, I, J, L, M, O



BFS Traversal

Step 28: Process node K

Legend

Complete

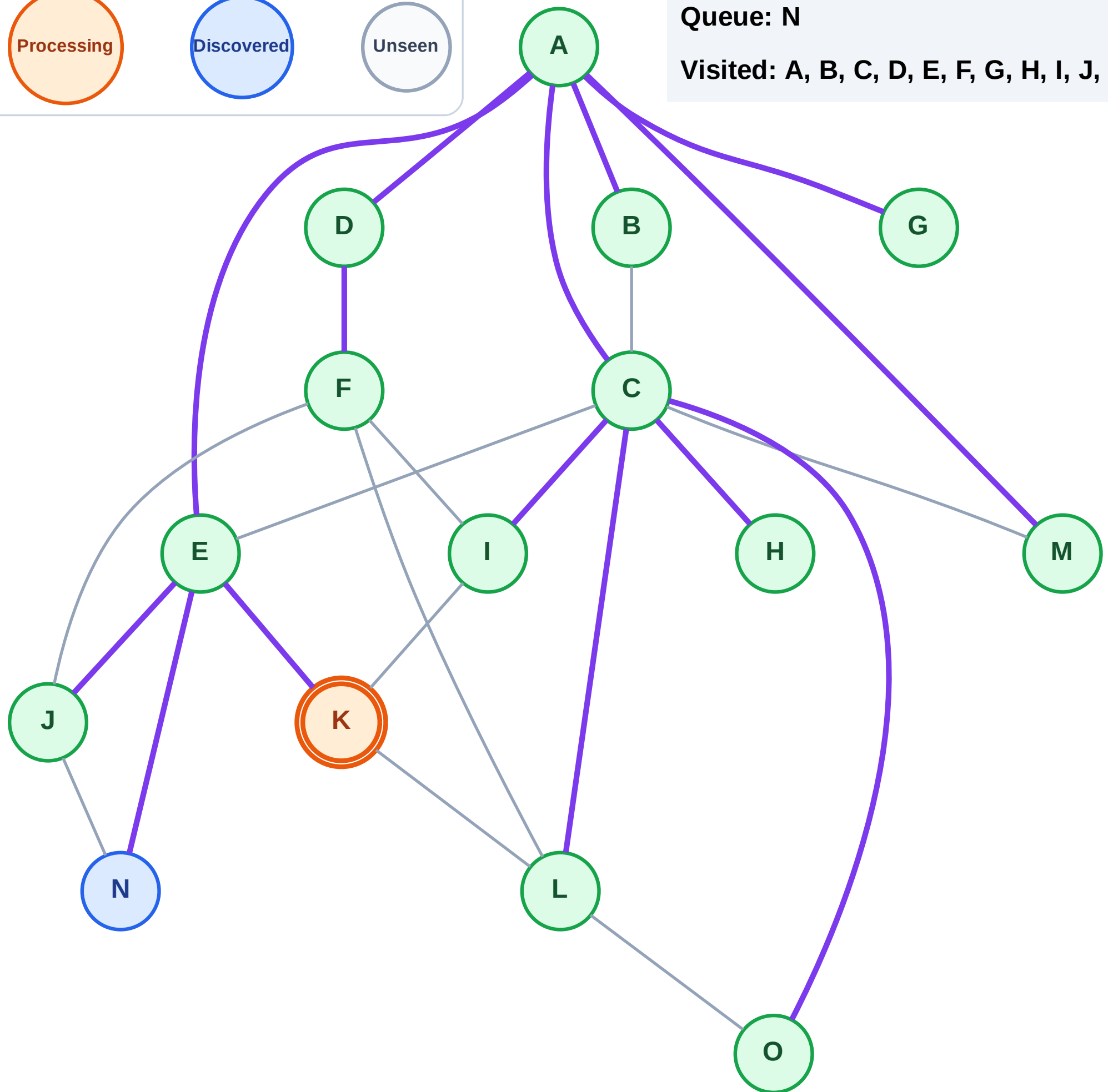
Processing

Discovered

Unseen

Queue: N

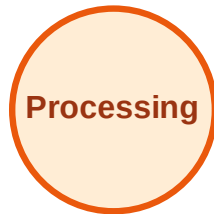
Visited: A, B, C, D, E, F, G, H, I, J, L, M, O



BFS Traversal

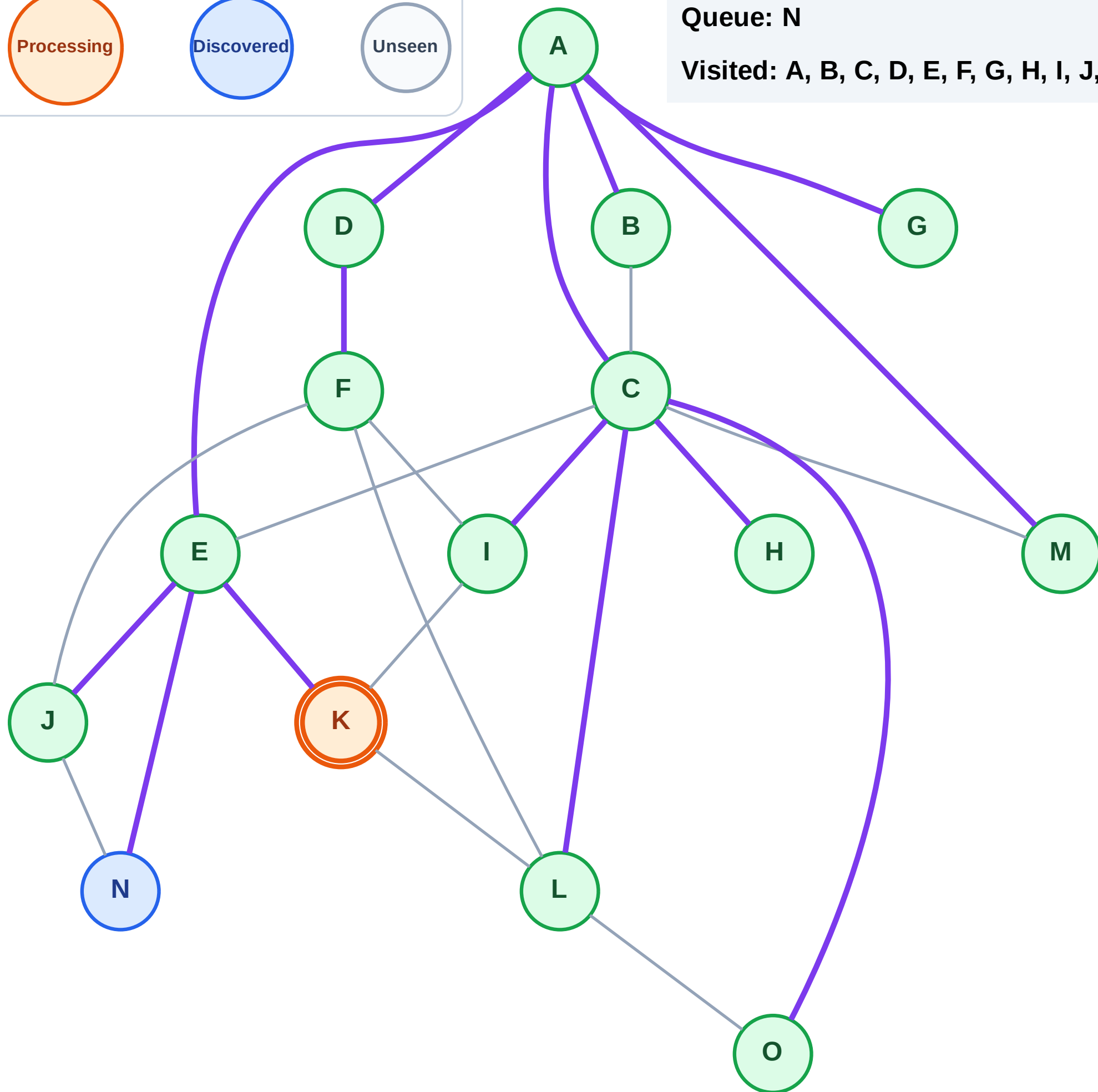
Step 29: TARGET NODE K REACHED! Path (2 hops): A → E → K

Legend



Queue: N

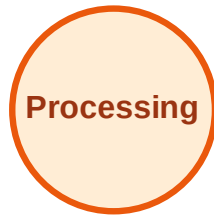
Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, O



DFS Traversal

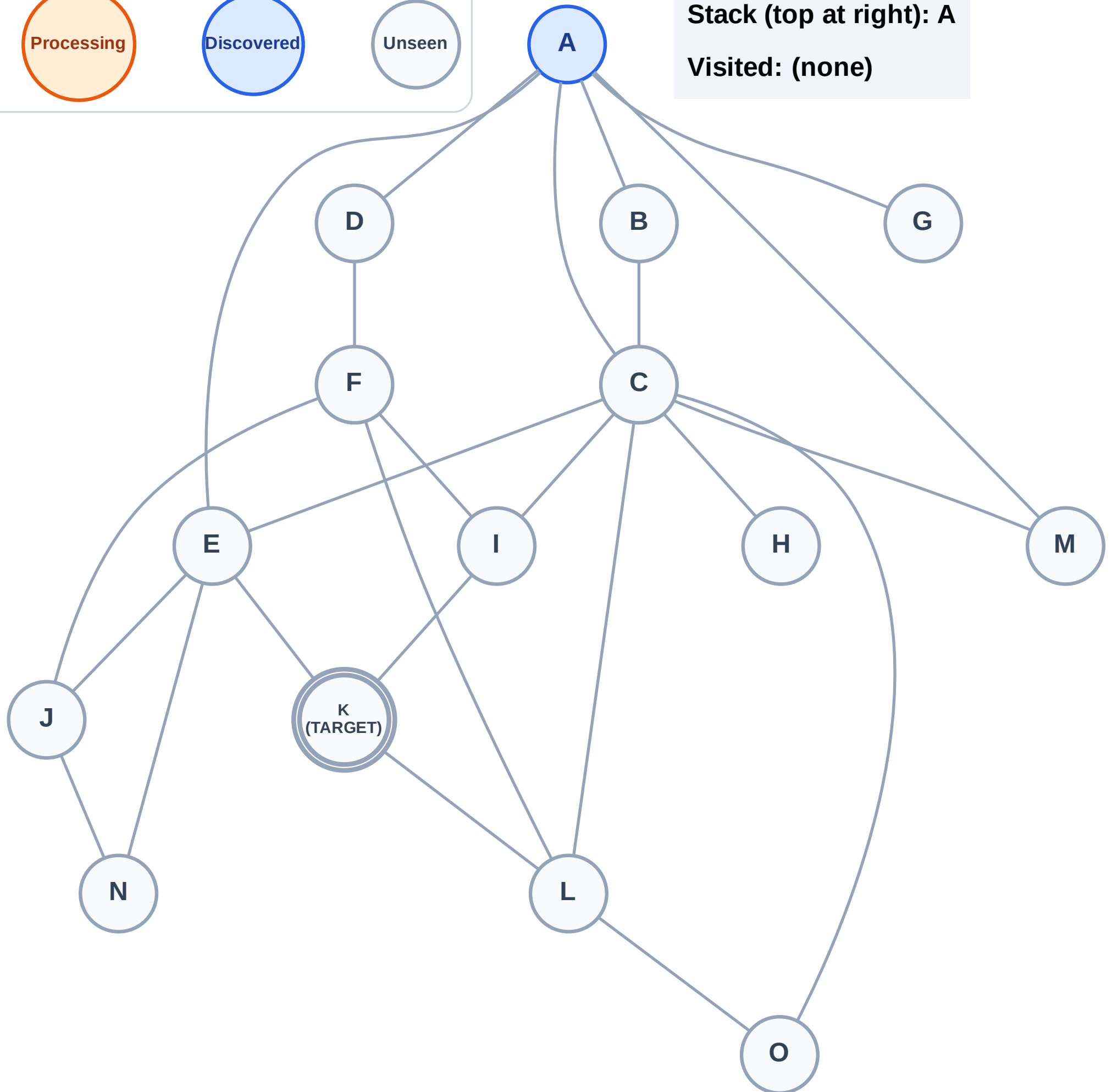
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

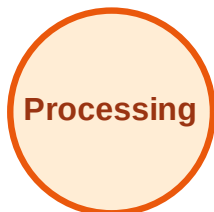
Visited: (none)



DFS Traversal

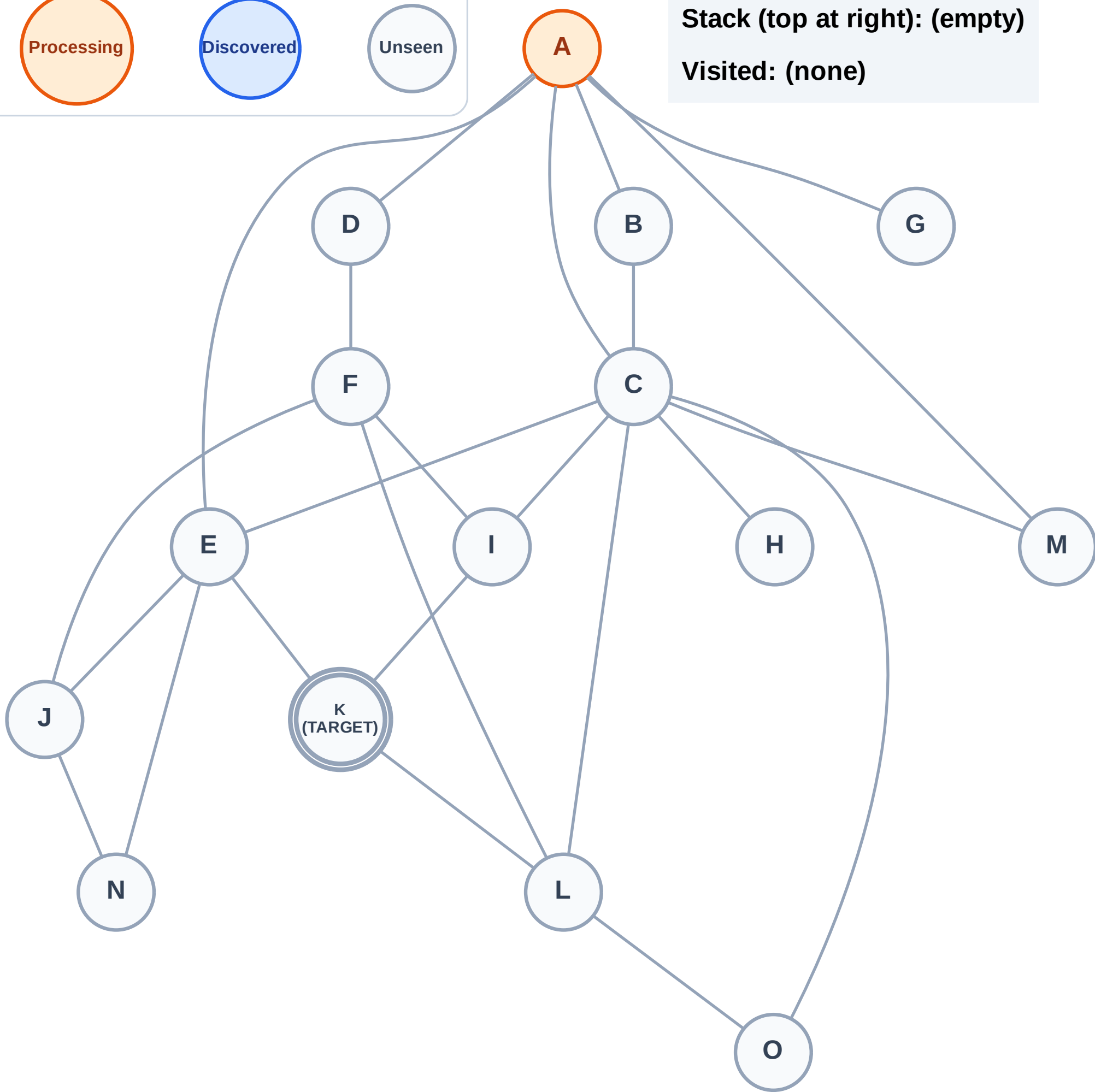
Step 2: Process node A

Legend



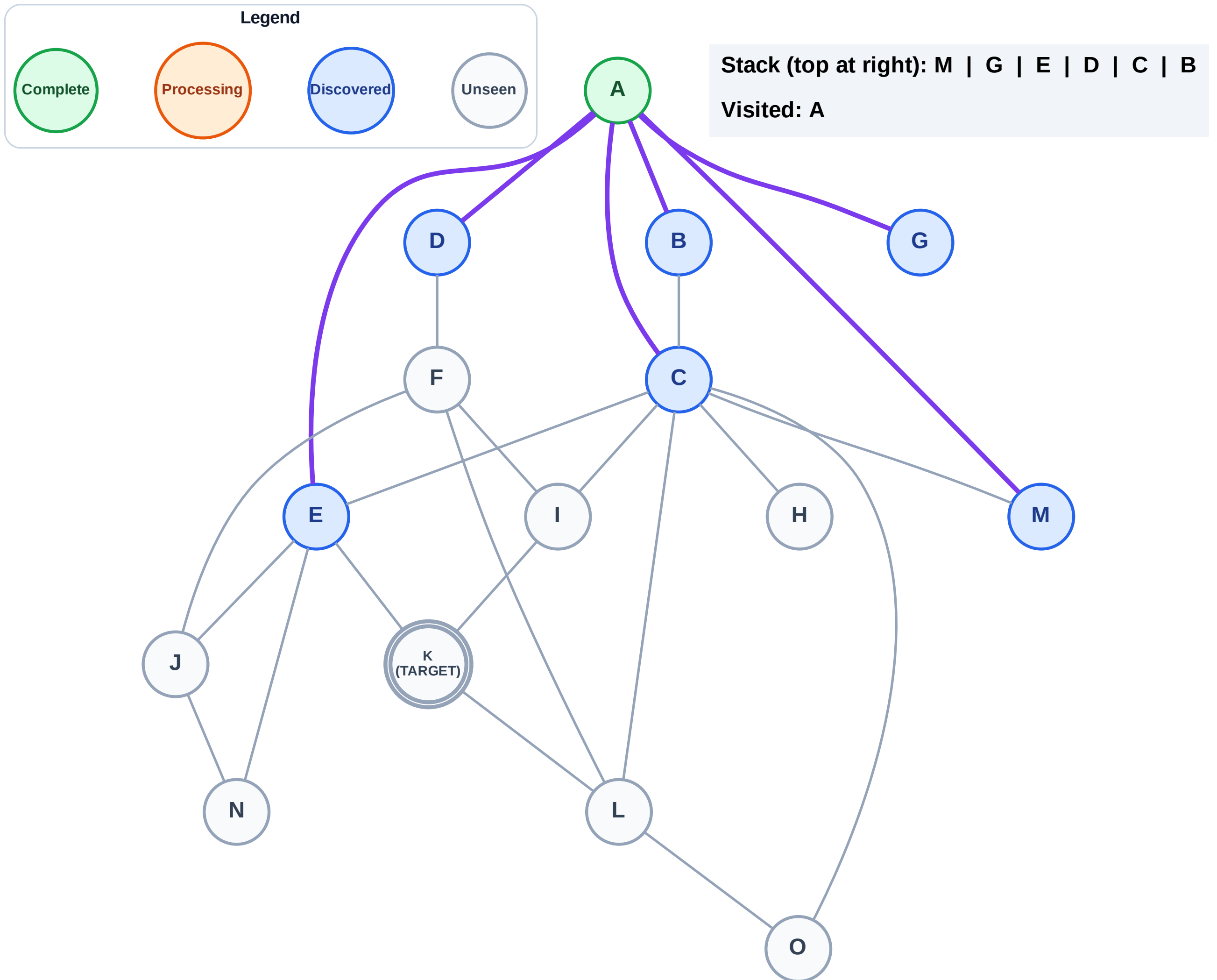
Stack (top at right): (empty)

Visited: (none)



Step 3: Discover M, G, E, D, C, B from A

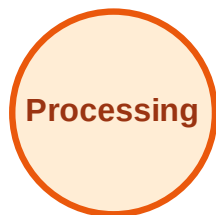
Step 3: Discover M, G, E, D, C, B from A



DFS Traversal

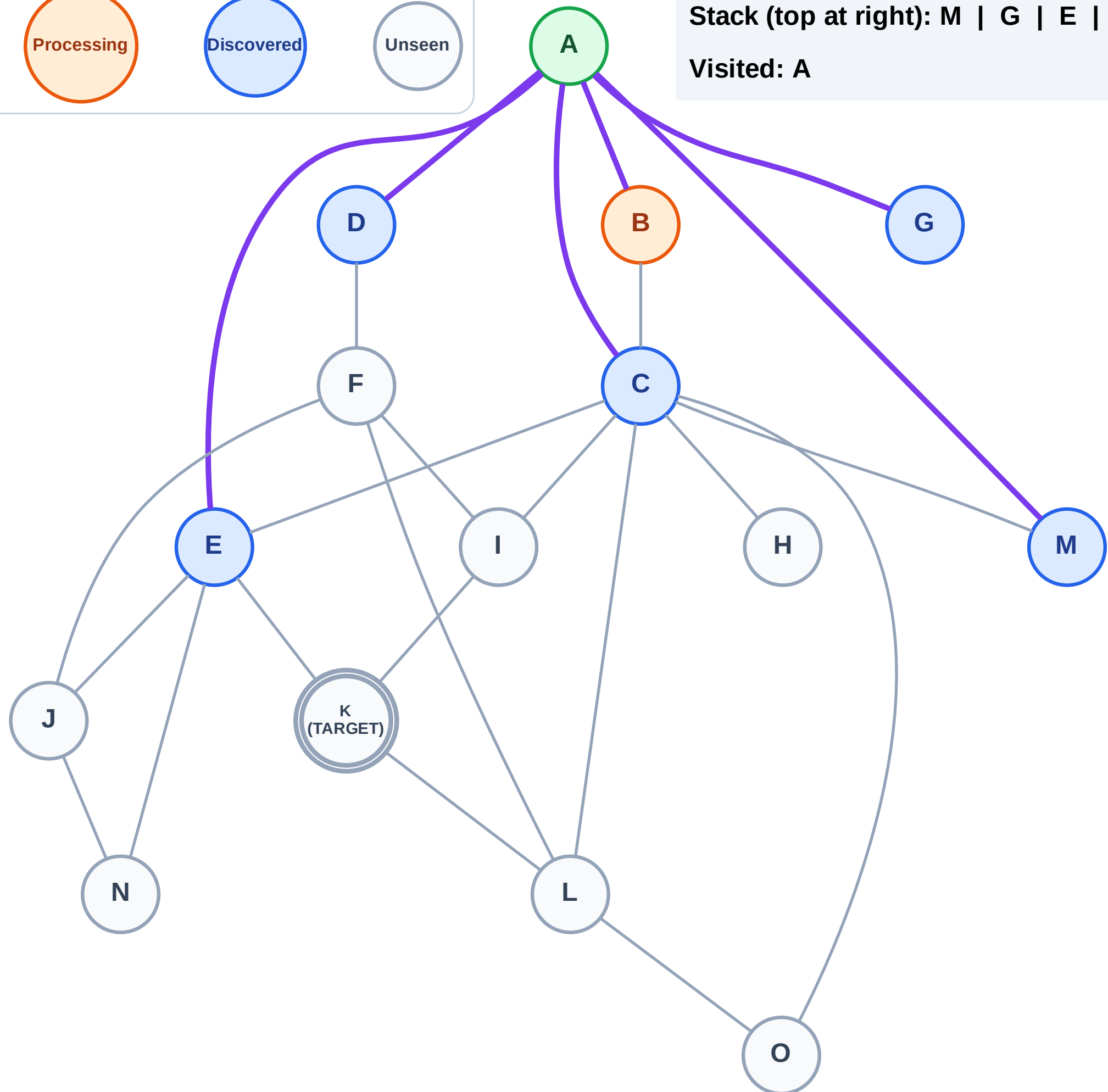
Step 4: Process node B

Legend



Stack (top at right): M | G | E | D | C

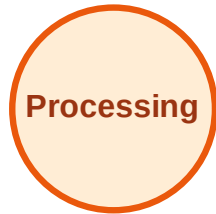
Visited: A



DFS Traversal

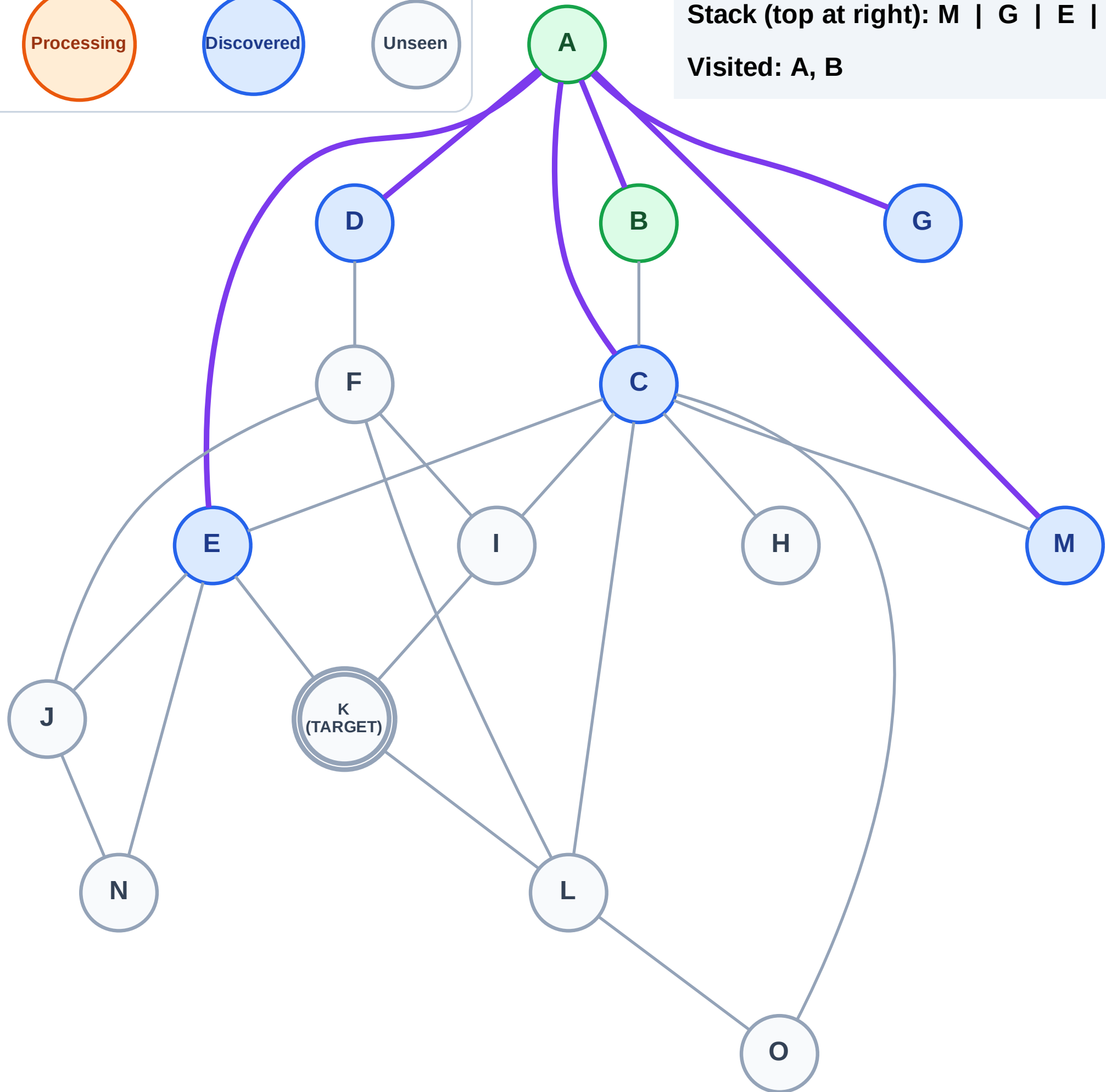
Step 5: No new neighbors from B

Legend



Stack (top at right): M | G | E | D | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

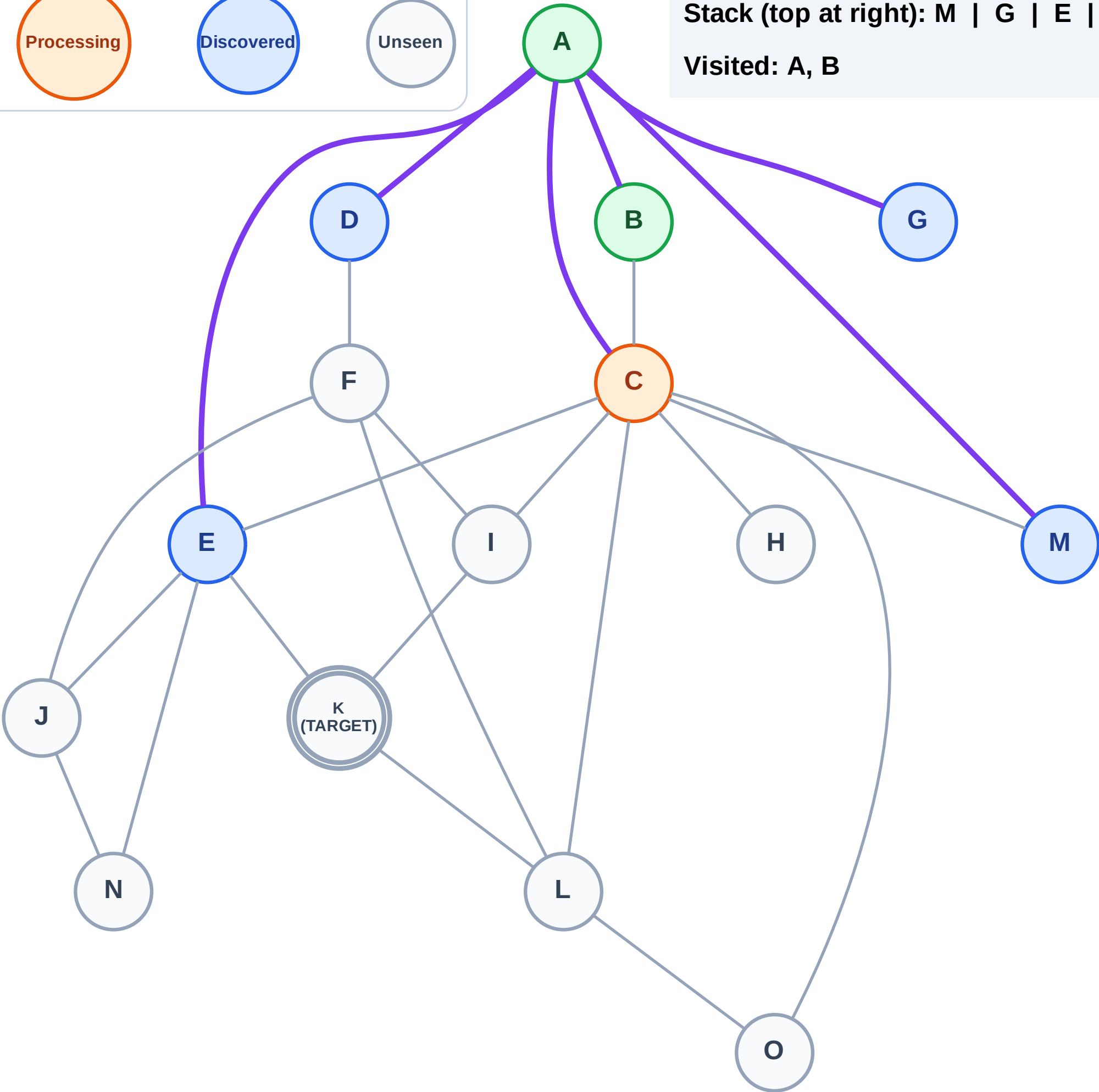
Processing

Discovered

Unseen

Stack (top at right): M | G | E | D

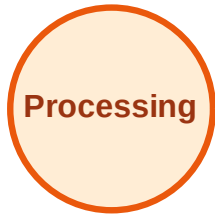
Visited: A, B



DFS Traversal

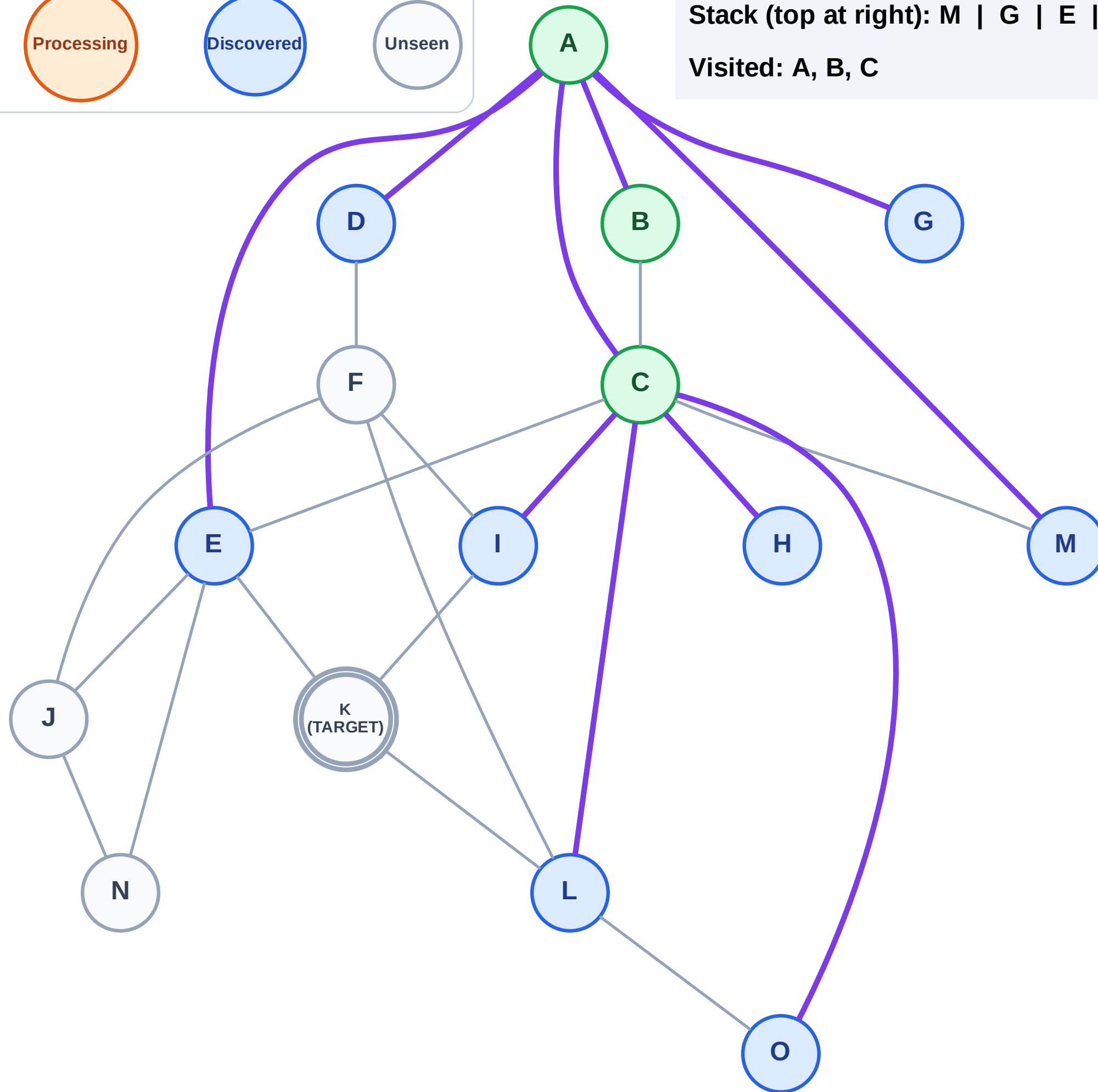
Step 7: Discover O, L, I, H from C

Legend



Stack (top at right): M | G | E | D | O | L | I | H

Visited: A, B, C



DFS Traversal

Step 8: Process node H

Legend

Complete

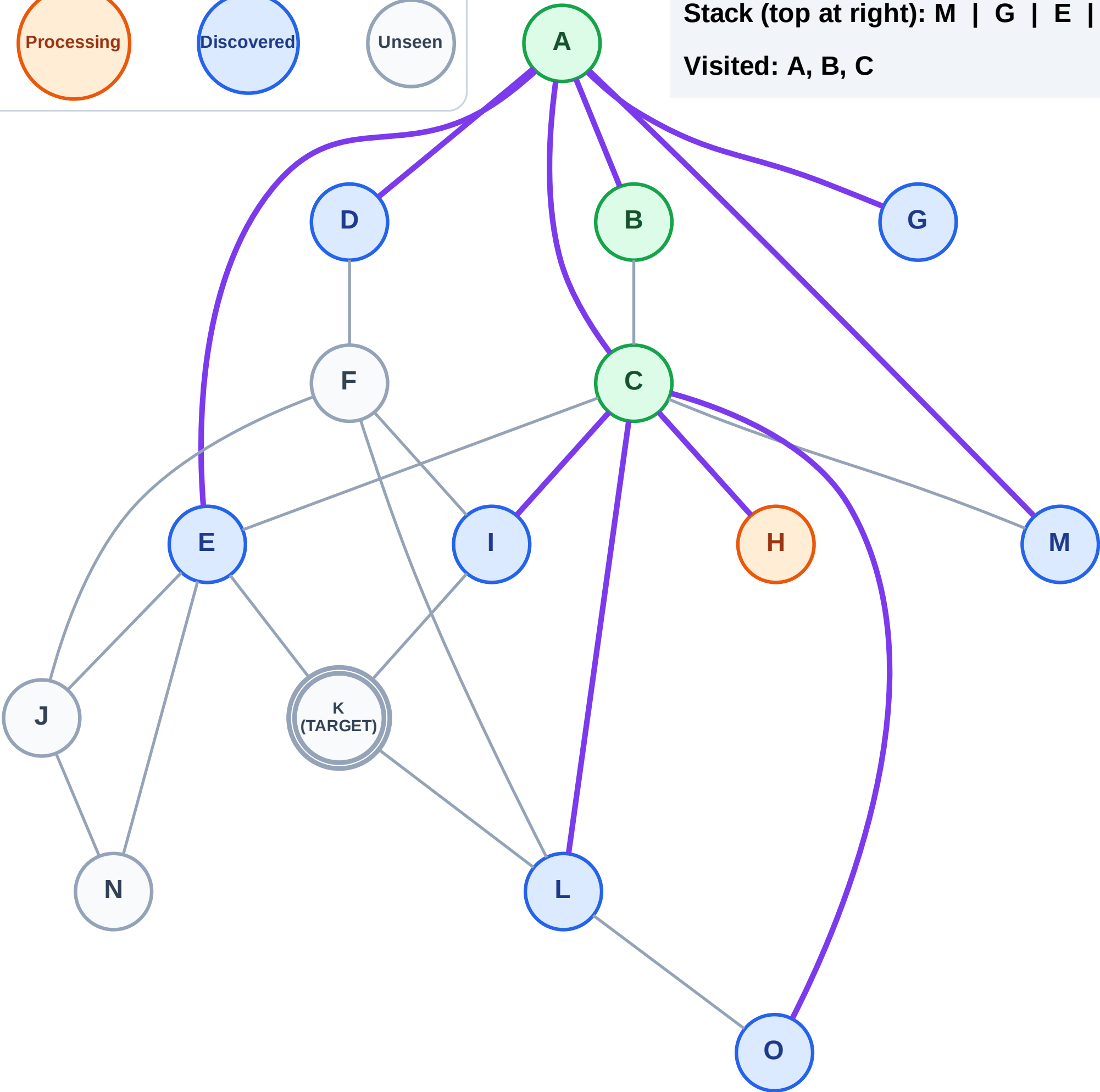
Processing

Discovered

Unseen

Stack (top at right): M | G | E | D | O | L | I

Visited: A, B, C



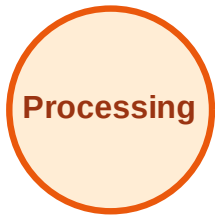
DFS Traversal

Step 9: No new neighbors from H

Legend



Complete



Processing



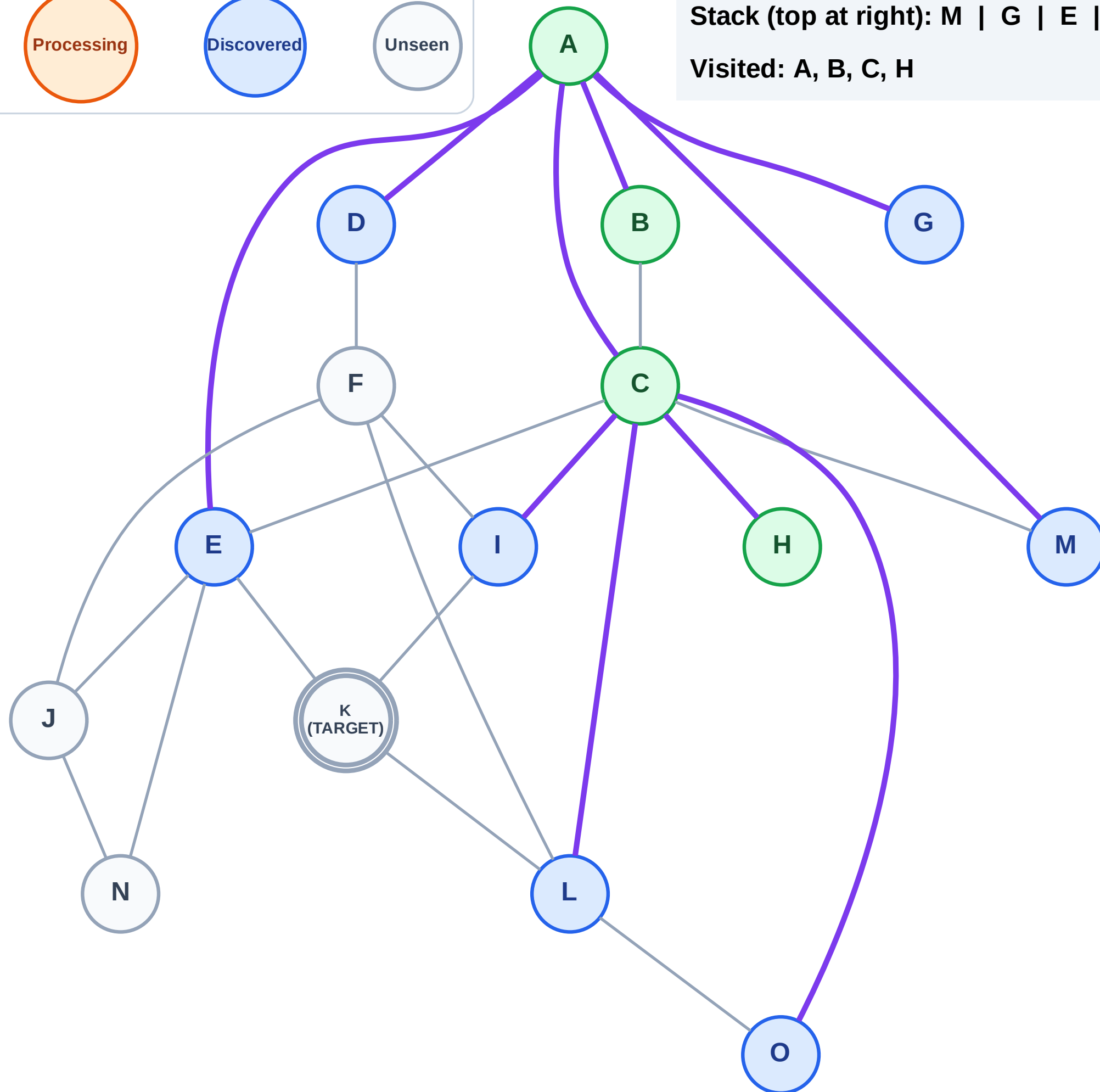
Discovered



Unseen

Stack (top at right): M | G | E | D | O | L | I

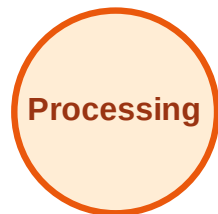
Visited: A, B, C, H



DFS Traversal

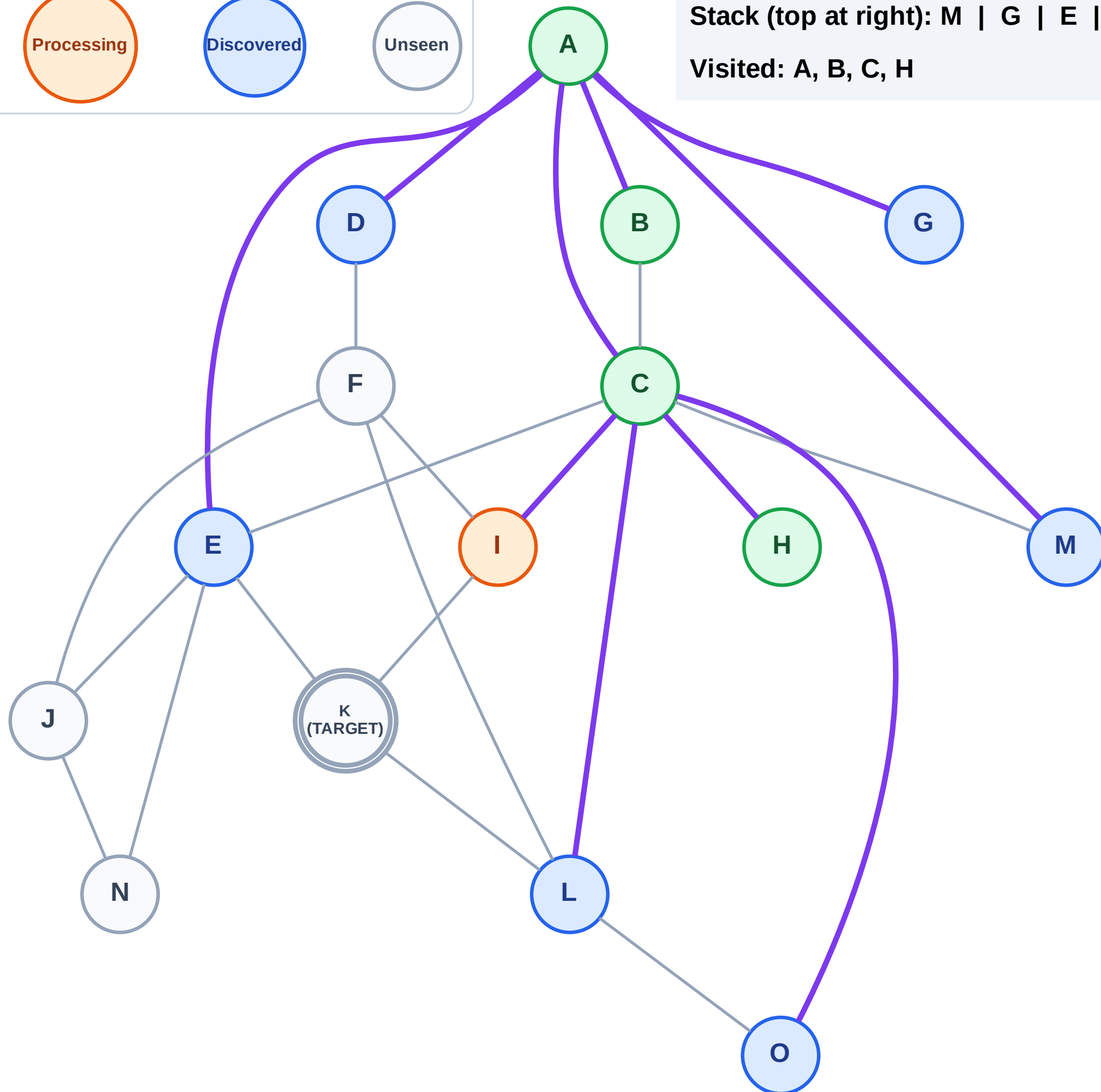
Step 10: Process node I

Legend



Stack (top at right): M | G | E | D | O | L

Visited: A, B, C, H



DFS Traversal

Step 11: Discover K, F from I

Legend

Complete

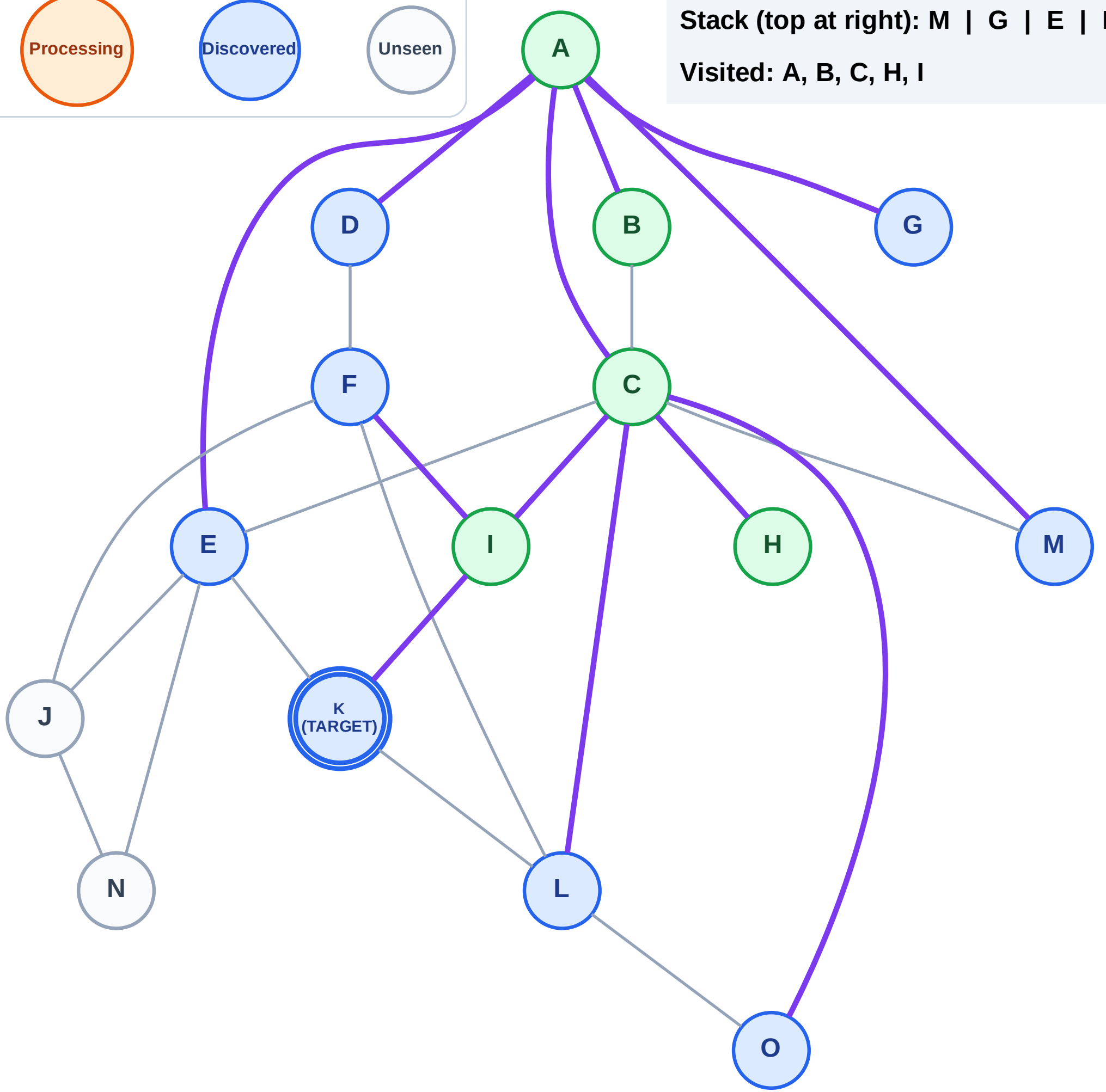
Processing

Discovered

Unseen

Stack (top at right): M | G | E | D | O | L | K | F

Visited: A, B, C, H, I



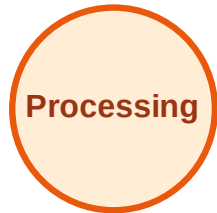
DFS Traversal

Step 12: Process node F

Legend



Complete



Processing



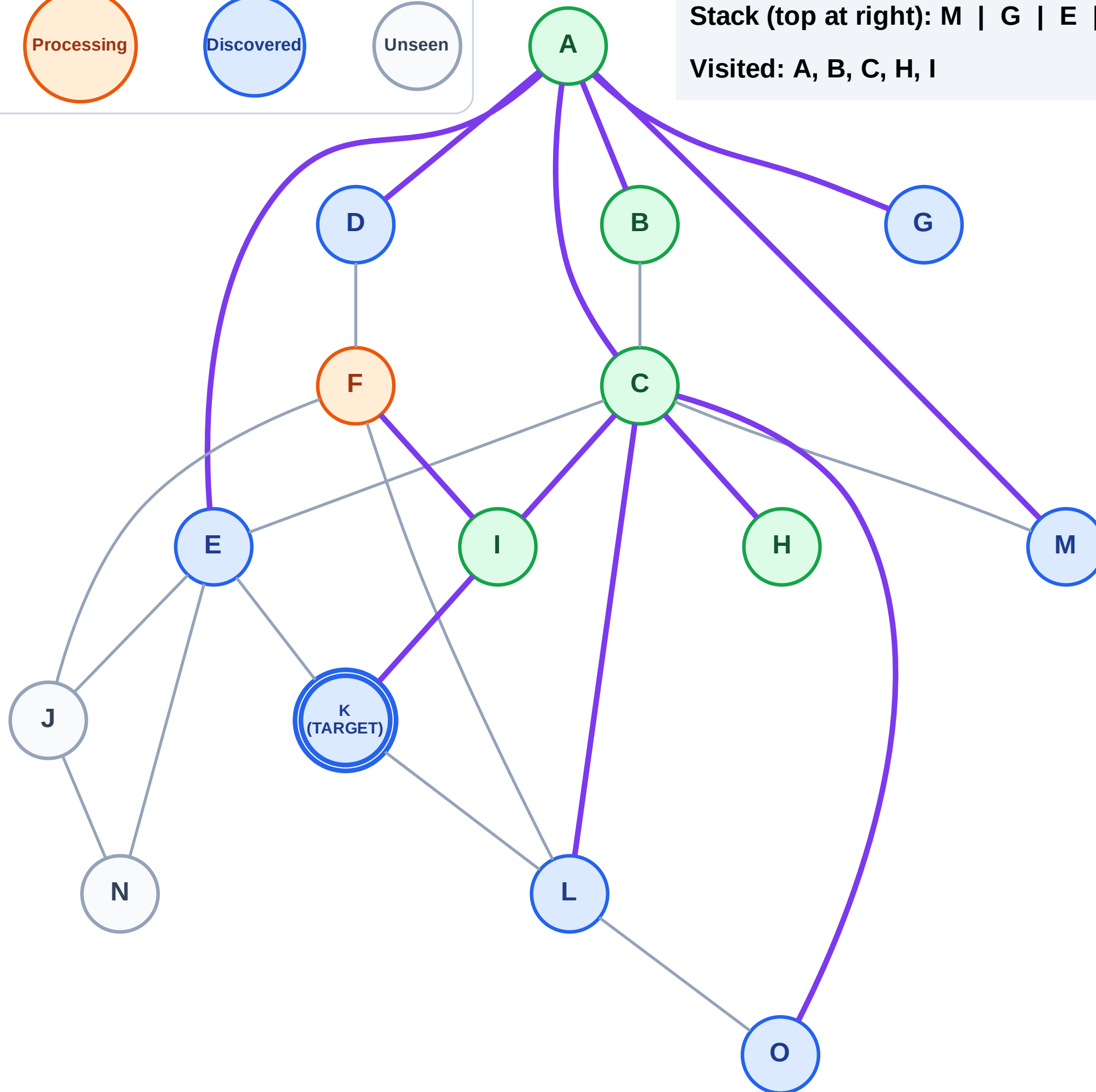
Discovered



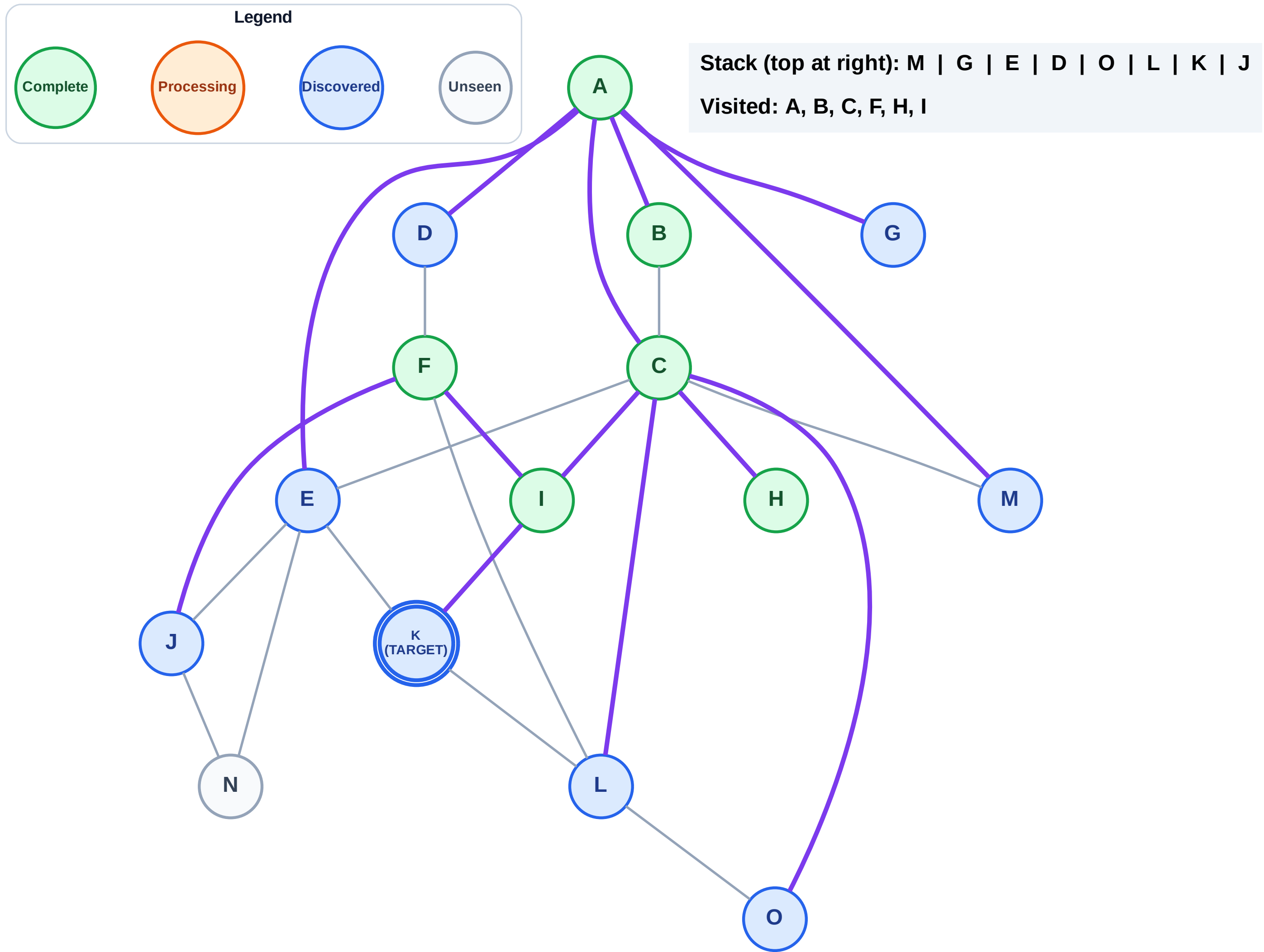
Unseen

Stack (top at right): M | G | E | D | O | L | K

Visited: A, B, C, H, I



Step 13: Discover J from F



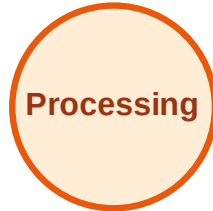
DFS Traversal

Step 14: Process node J

Legend



Complete



Processing



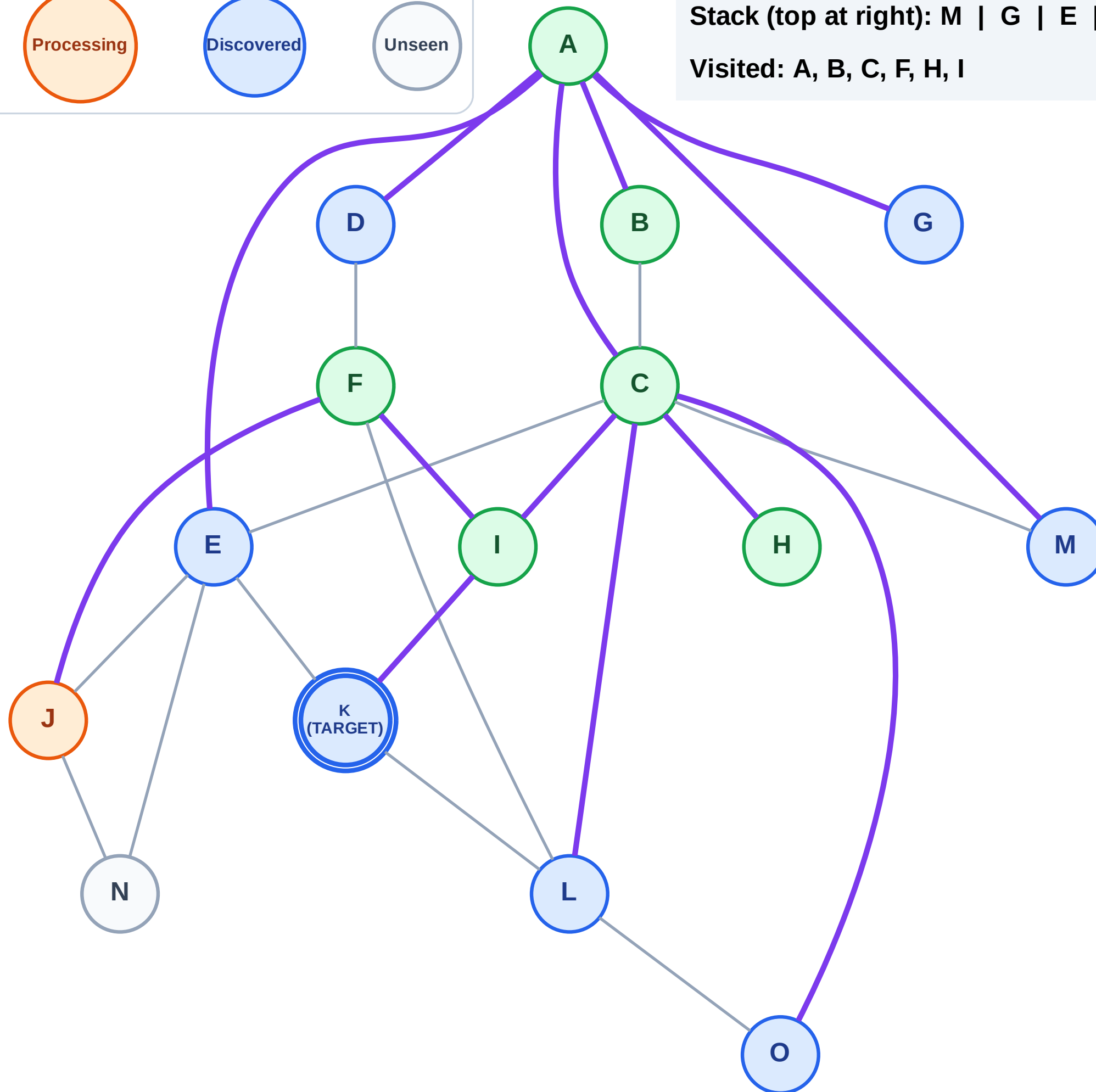
Discovered



Unseen

Stack (top at right): M | G | E | D | O | L | K

Visited: A, B, C, F, H, I



DFS Traversal

Step 15: Discover N from J

Legend

Complete

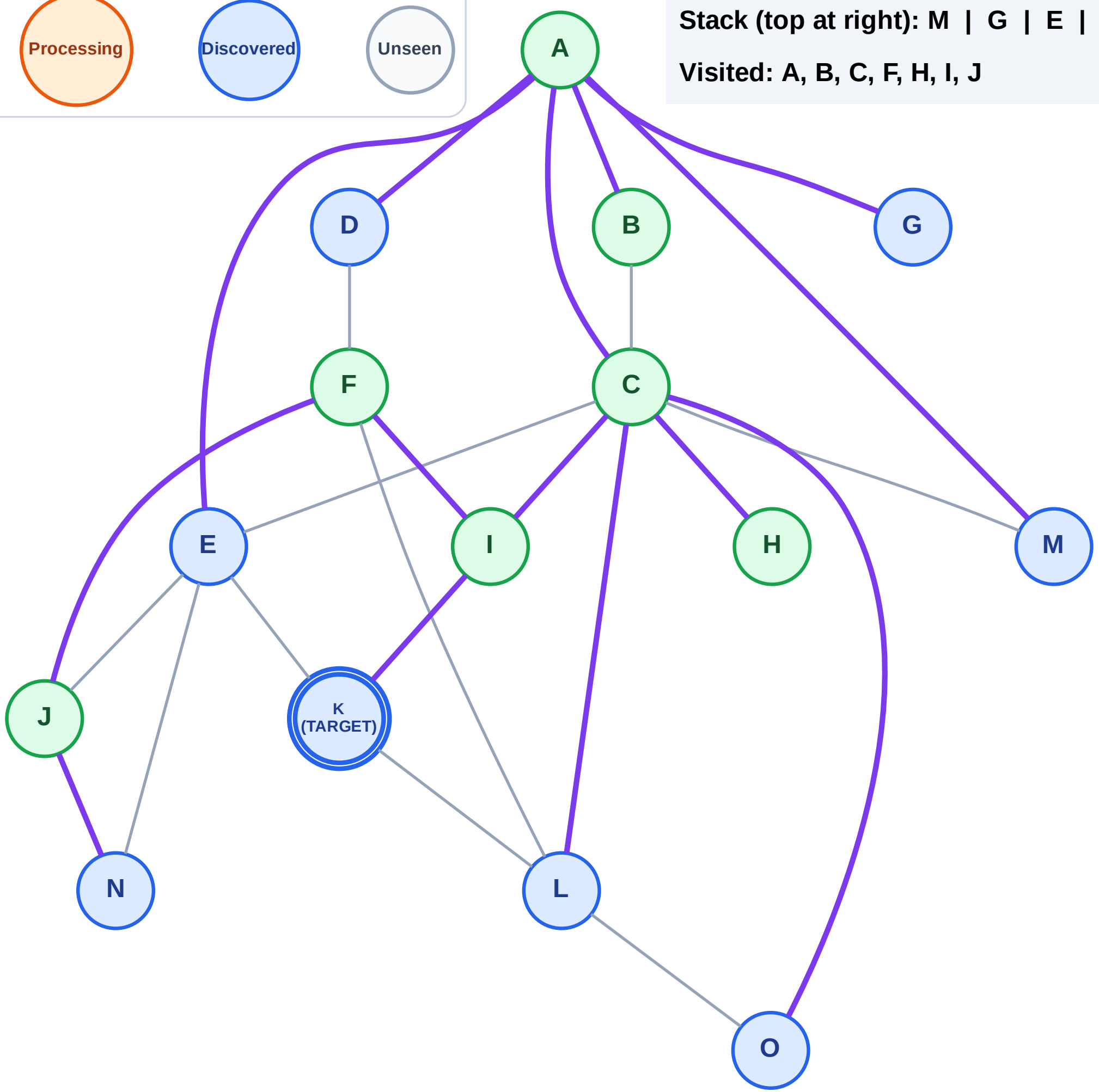
Processing

Discovered

Unseen

Stack (top at right): M | G | E | D | O | L | K | N

Visited: A, B, C, F, H, I, J



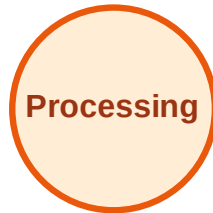
DFS Traversal

Step 16: Process node N

Legend



Complete



Processing



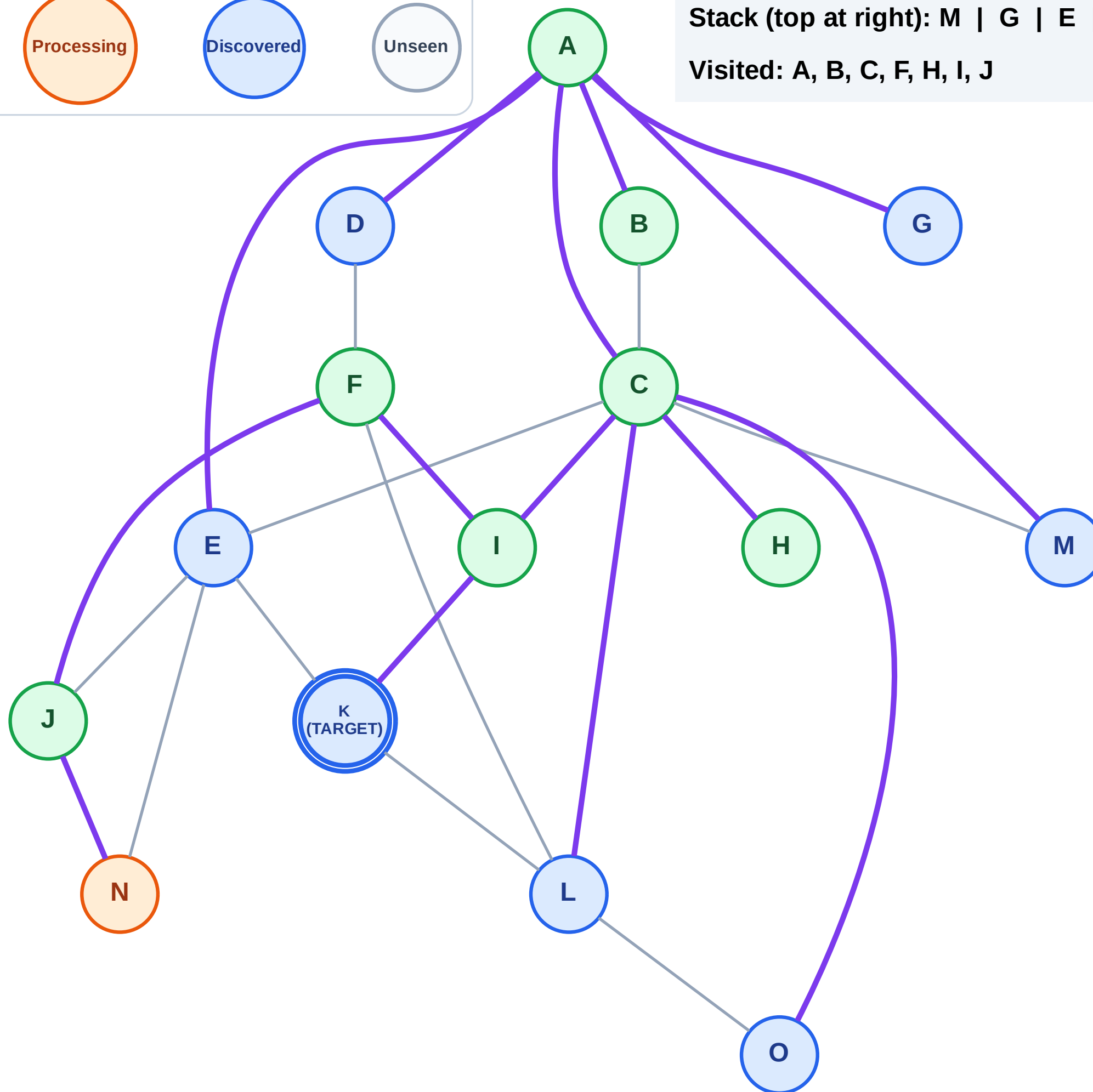
Discovered



Unseen

Stack (top at right): M | G | E | D | O | L | K

Visited: A, B, C, F, H, I, J



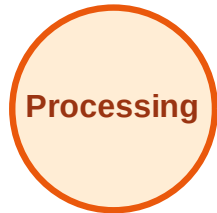
DFS Traversal

Step 17: No new neighbors from N

Legend



Complete



Processing



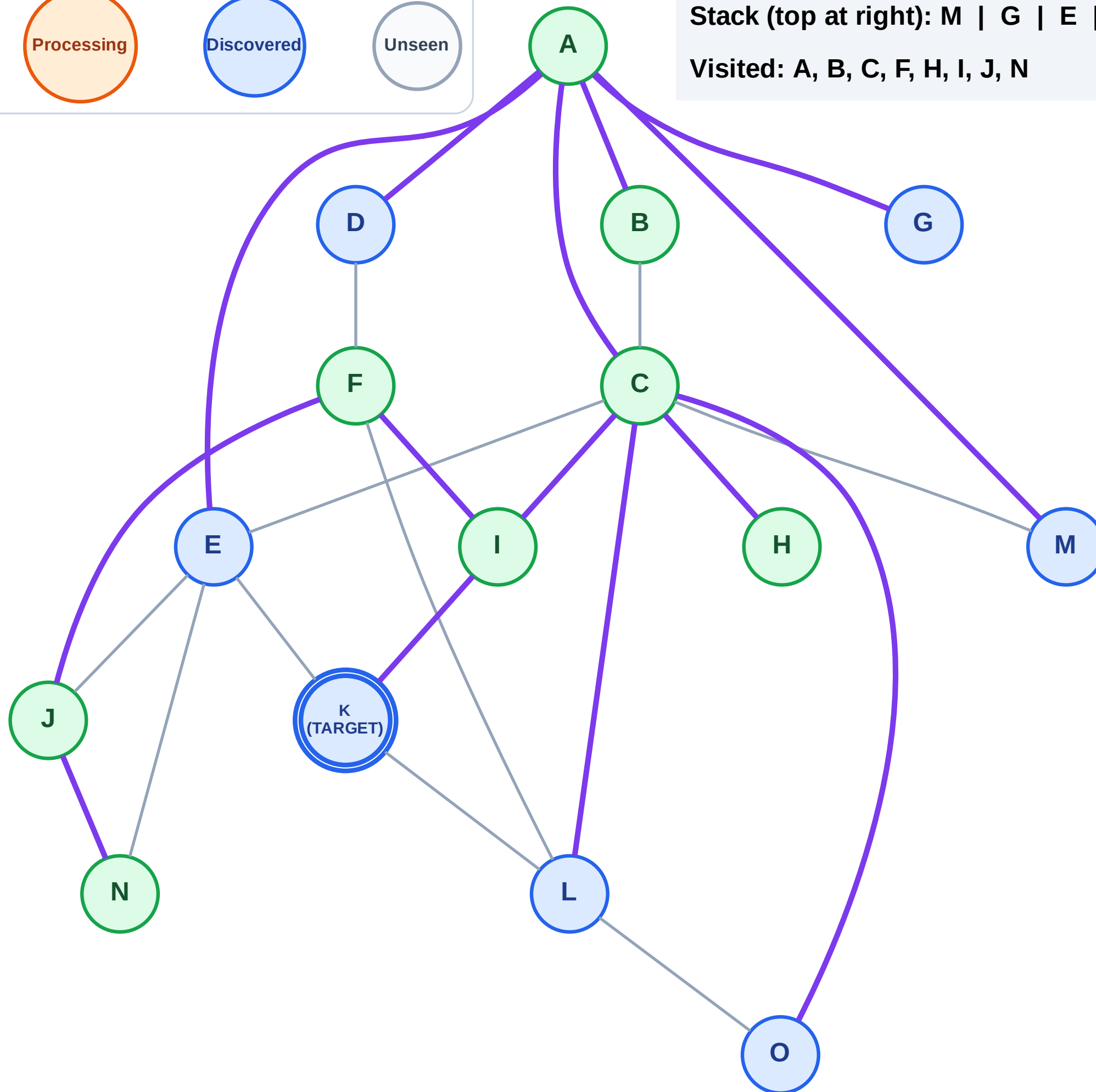
Discovered



Unseen

Stack (top at right): M | G | E | D | O | L | K

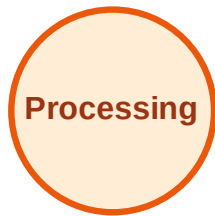
Visited: A, B, C, F, H, I, J, N



DFS Traversal

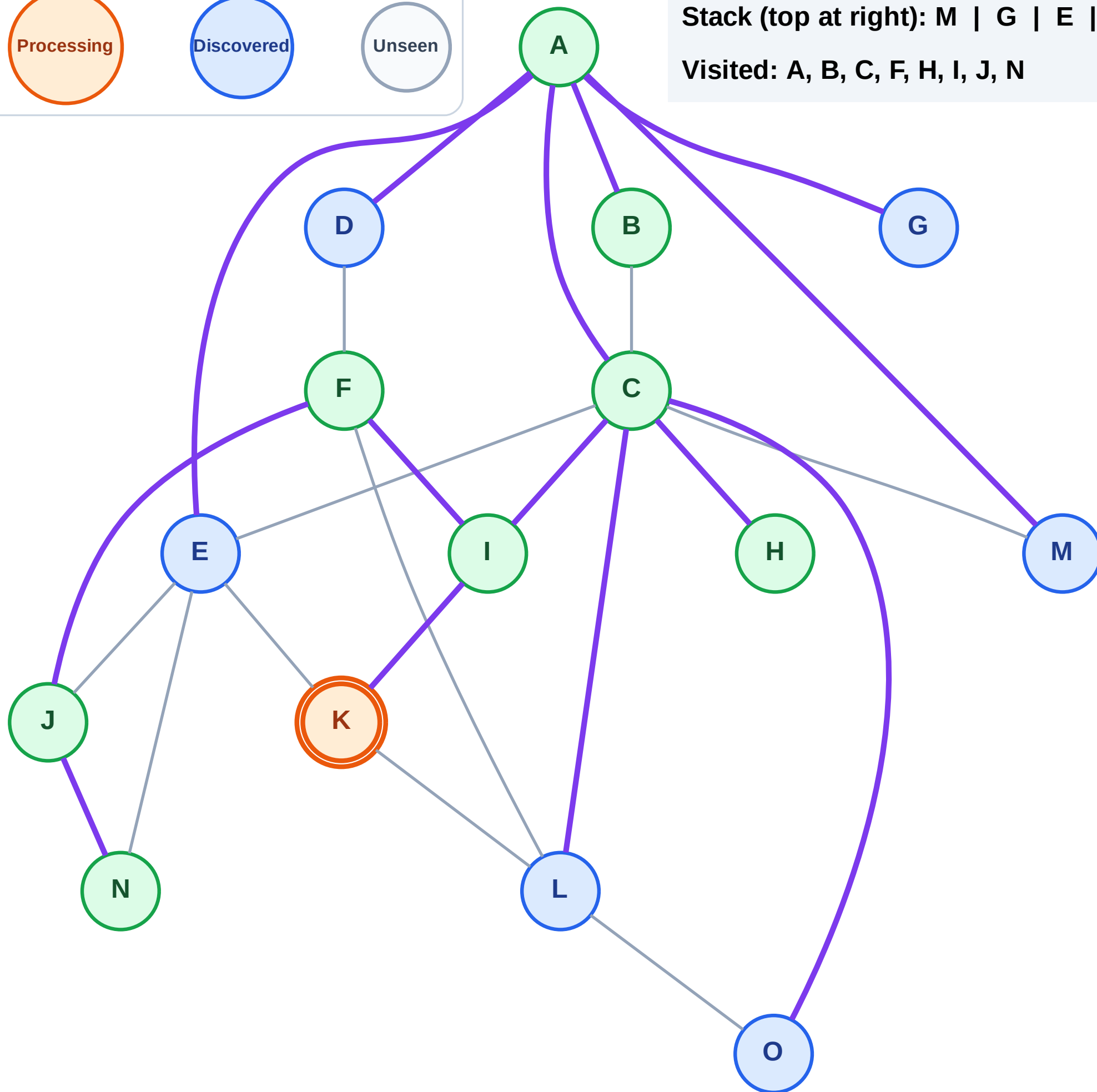
Step 18: Process node K

Legend



Stack (top at right): M | G | E | D | O | L

Visited: A, B, C, F, H, I, J, N



DFS Traversal

Step 19: TARGET NODE K REACHED! Path (3 hops): A → C → I → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): M | G | E | D | O | L

Visited: A, B, C, F, H, I, J, K, N

